02 Redox and Equivalent Concept

Introduction of Redox Reaction:

Many Chemical reactions involved transfer of electrons from one chemical substance to another. These electron-transfer reactions are termed as **Oxidation-Reduction** or **Redox reactions.**

or

Those reactions which involves oxidation and reduction both simultaneously are known as oxidation and reduction or Redox Reaction.

or

Those reaction which involves increase in oxidation number and decrease in oxidation number simultaneously are known as Oxidation & Reduction or Redox Reaction.

Oxidation and Reduction:

There are four concepts for oxidation and reduction reactions.

(A) Classical concept
(B) Modern concept
(C) Valency concept
(D) Oxidation number concept.

Classical concept :-

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According to this concept, oxidation and reduction can be explained as :

(a) Oxidation :- Oxidation is a process which involves.

(i) Addition of Oxygen :

Example :

$2Mg + O_2 \longrightarrow 2MgO$ (Oxidation of Magnesium)

$C + O_2 \longrightarrow CO_2$ (Oxidation of Carbon)

$Na_2SO_3 + H_2O_2 \longrightarrow Na_2SO_4 + H_2O$ (Oxidation of Sodium sulphite)

(ii) Removal of Hydrogen :

Example :

$H_2S + Cl_2 \longrightarrow 2HCl + S$ (Oxidation of Hydrogen sulphide)

$4HI + O_2 \longrightarrow 2H_2O + 2I_2$ (Oxidation of Hydrogen iodide)

(iii) Addition of Electronegative element :

Example :

$Fe + S \longrightarrow FeS$ (Oxidation of iron)

$SnCl_2 + Cl_2 \longrightarrow SnCl_4$ (Oxidation of stannous chloride)

$2FeCl_2 + Cl_2 \longrightarrow 2FeCl_3$ (Oxidation of ferrous chloride)

(iv) Removal of Electropositive element :

Example :

$2NaI + H_2O_2 \longrightarrow 2NaOH + I_2$ (Oxidation of NaI)

$2KI + Cl_2 \longrightarrow 2KCl + I_2$ (Oxidation of KI)

$2K_2MnO_4 + Cl_2 \longrightarrow 2KCl + 2KMnO_4$ (Oxidation of Potassium manganate)

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(v) Increment in oxidation state of Electropositive element :

Example :

$SnCl_2 + Cl_2 \longrightarrow SnCl_4$

$Sn^{2\oplus} \longrightarrow Sn^{4\oplus}$

(b) Reduction : Reduction is a process which involves

(i) Removal of Oxygen :

Example :

$CuO + C \longrightarrow Cu + CO$ (Reduction of cupric oxide)

$\underset{\text{Steam}}{H_2O} + \underset{\text{Coke}}{C} \longrightarrow \underbrace{CO + H_2}_{\text{Water gas}}$ (Reduction of water)

$Fe_3O_4 + 4H_2 \longrightarrow 3Fe + 4H_2O$ (Reduction of Fe_3O_4)

(ii) Addition of Hydrogen

Example :

$N_2 + 3H_2 \longrightarrow 2NH_3$ (Reduction of Nitrogen)

$Cl_2 + H_2 \longrightarrow 2HCl$ (Reduction of Chlorine)

$S + H_2 \longrightarrow H_2S$ (Reduction of Sulphur)

(iii) Removal of Electronegative element

Example :

$2HgCl_2 + SnCl_2 \longrightarrow Hg_2Cl_2 + SnCl_4$ (Reduction of mercuric chloride)

$2FeCl_3 + H_2 \longrightarrow 2FeCl_2 + 2HCl$ (Reduction of ferric chloride)

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(iv) Addition of Electropositive element

Example :

$HgCl_2 + Hg \longrightarrow Hg_2Cl_2$ (Reduction of mercuric chloride)

$CuCl_2 + Cu \longrightarrow Cu_2Cl_2$ (Reduction of cupric chloride)

$2HgCl_2 + SnCl_2 \longrightarrow Hg_2Cl_2 + SnCl_4$ (Reduction of mercuric chloride)

(v) Decrement in oxidation state of Electropositive element

Example :

$CuSO_4 + Fe \longrightarrow FeSO_4 + Cu$ (Reduction of $CuSO_4$)

$Cu^{2\oplus} \quad\quad Cu^0$

$FeCl_3 + H_2S \longrightarrow FeCl_2 + 2HCl + S$ (Reduction of $FeCl_3$)

$Fe^{3\oplus} \quad\quad Fe^{2\oplus}$

Modern concept or Electronic concept:

(a) Oxidation : According to this concept the process which involves the loss of one or more electrons from an atom or an ion or molecule is called oxidation (de-electronation).

Example :

$M \longrightarrow M^{n\oplus} + ne^{\ominus}$

$Mg \longrightarrow Mg^{2\oplus} + 2e^{\ominus}$

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$M^{n_1\oplus} \longrightarrow M^{n_2\oplus} + (n_2 - n_1)e^{\ominus}$

$Fe^{2\oplus} \longrightarrow Fe^{3\oplus} + e^{\ominus}$

(ferrous) (ferric)

$A^{n\ominus} \longrightarrow A + ne^{\ominus}$

$2Cl^{\ominus} \longrightarrow Cl_2 + 2e^{\ominus}$

$A^{n_1\ominus} \longrightarrow A^{n_2\ominus} + (n_1 - n_2)\, e^{\ominus}$

$[Fe(CN)_6]^{4\ominus} \longrightarrow [Fe(CN)_6]^{3\ominus} + e^{\ominus}$

(b) Reduction : According to this concept, the process which involves gain of one or more electrons by an atom or on ion or molecule is called reduction (electronation).

Example :

$M^{n_2\oplus} + (n_2 - n_1)e^{\ominus} \longrightarrow M^{n_1\oplus}$

$Sn^{4\oplus} + 2e^{\ominus} \longrightarrow Sn^{2\oplus}$

(Stannic ion) (Stannous ion)

$A + ne^{\ominus} \longrightarrow A^{n\ominus}$

$Cl + e^{\ominus} \longrightarrow Cl^{\ominus}$

$A^{n_1\ominus} + (n_2 - n_1)\, e^{\ominus} \longrightarrow A^{n_2\ominus}$

$MnO_4^{\ominus} + e^{\ominus} \longrightarrow MnO_4^{2\ominus}$

$[Fe(CN)_6]^{3\ominus} + e^{\ominus} \longrightarrow [Fe(CN)_6]^{4\ominus}$

Oxidation Number:

Definition : It represents the number of electron gained or lost by an atom when it changes in to a compound from a free state.

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Or

[It is the charge developed on an atom when atom is in combination.]

(a) If electron is gained by an atom in the formation of compound, oxidation number is given (–) ve sign.

(b) If electron is lost by an atom in the formation of compound, oxidation number is given (+) ve sign.

(c) It represents the real charge in case of ionic compounds and represents the imaginary charge in case of covalent compounds.

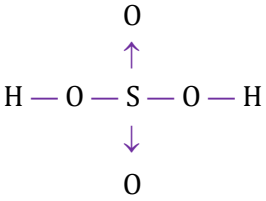
(d) [Maximum oxidation no. of an element is equal to group no. which be belongs in the periodic table.]

(e) Minimum oxidation no. of an element is equal to (group number – 8.)

Oxidation no. for Coordinate bond :-

(a) When coordinate bond is formed in between less electronegative element and high electronegative element then the $e^{\ominus}$ donor element show + 2 oxidation number where as $e^{\ominus}$ acceptor element show – 2 oxidation number In this type of bond or bonded compounds, Electronegativity of acceptor > Electronegativity of donor.

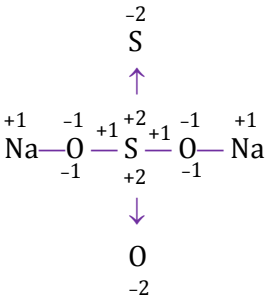
For example, in H_2SO_4.



Here 'S' is less electronegative than O therefore, oxidation number of S = +2 and oxidation number of O = – 2 (due to single dative bond).

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(b) When coordinate bond is formed between the two same electronegative elements then the $e^{\ominus}$ donor element show + 2 oxidation number whereas $e^{\ominus}$ acceptor element show – 2 oxidation number. In this type of bond or bonded compound, E. N. of acceptor = E. N. of donor.
For example : In $Na_2S_2O_3$



Here oxidation number of 'S' is + 2. Because it is $e^{\ominus}$ pair donor and the other 'S' is – 2, Because it is $e^{\ominus}$ pair acceptor. (Due to dative bond)

(c) When coordinate bond is formed in between high electronegative element to low electronegative element then change in oxidation number will be shown by both the elements, which are bonded by coordinate bond e.g. HCN.

$H_{+1}—_{-1}N^0 \longrightarrow C^0$

Oxidation State:

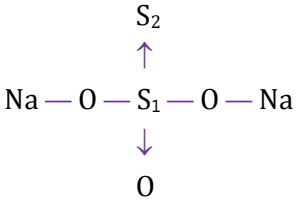
Oxidation state of an atom is defined as oxidation number per atom for all practical purposes. Oxidation state is often expressed as oxidation number.

The rules to derive oxidation number or oxidation state :-

(a) The oxidation state of an element in its free state is zero. Example oxidation state of Na, Cu, I, Cl, O etc. are zero.
(b) Sum of oxidation state of all the atoms in a neutral molecule is zero.
(c) Sum of oxidation state of all the atoms in a complex ion is equal to number of charges present on it.
(d) In complex compounds, oxidation state of some neutral molecules (ligands) is zero. Example CO, NO, NH_3, H_2O.
(e) Generally, oxidation state of oxygen is – 2 but in H_2O_2 it is – 1 & in OF_2 it is +2.
(f) Generally, oxidation state of Hydrogen is + 1 but in metallic hydrides it is –1.
(g) Generally, oxidation state of Halogen atoms is – 1 but in interhalogen compounds it changes.
(h) Generally, oxidation state of alkali metals is + 1 and that of alkaline earth metals is +2.
(i) Oxidation state of transition elements varies from compound to compound. Mn has oxidation state from + 1 to + 7.
$Mn_2O \rightarrow +1$, $MnO \rightarrow +2$, $Mn_3O_4 \rightarrow +8/3$, $MnO_2 \rightarrow +4$, $Mn_2O_5 \rightarrow +5$, $MnO_4^{2\ominus} \rightarrow +6$, $MnO_4^{\ominus} \rightarrow +7$.
(j) Oxidation state of an atom may be fractional, negative, zero as well as positive.

Note :

Sometimes same atom in a compound has different oxidation state for example, structures of $Na_2S_2O_3$ is



Here S_1 and S_2 both are sulphur atoms but they have different oxidation state oxidation state of S_1 = +6 Oxidation state of S_2 = – 2 (It is accepting two electrons from S_1) Average oxidation state of sulphur atom is +2

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Oxidation State as a periodic property:

Oxidation state of an atom depends upon the electronic configuration of an atom i.e. why it is a periodic property.

(a) I A group element (alkali metals) show + 1 oxidation state.

(b) II A group element (alkaline earth metals) show + 2 oxidation state

(c) The maximum normal oxidation state show by III A group elements is +3. These elements also show +2 to +1 oxidation state as well.

(d) Elements of IV A group show their maximum and minimum oxidation states as +4 and –4 respectively.

(e) Non-metals shows number of oxidation states, the relation between maximum and minimum oxidation states for non-metals is equal to (maximum O.S. – minimum O.S. = 8).

For example, sulphur has maximum oxidation number +6 as being in VI A group element.

Illustration 1

Calculate oxidation number of underlined element $Na_2\underline{S}_2O_3$:

Solution:

Let oxidation number of S-atom is x. Now work accordingly with the rules given before.

$(+1) \times 2 + (x) \times 2 + (-2) \times 3 = 0$

$x = +2$

Illustration 2

$Na_2\underline{S}_4O_6$:

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Solution:

Let oxidation number of S-atom is x

$(+1) \times 2 + (x) \times 2 + (-2) \times 6 = 0$

$x = +2.5$

- It's important to note here that $Na_2S_2O_3$ have two S-atom and there are four S-atom in $Na_2S_4O_6$ but sulphur atom in both the compound have +2 or +2.5 oxidation number, it is the average charge (O.N.) which reside on each sulphur atom therefore we should work to calculate the individual oxidation number of each sulphur atom in this compound.

Illustration 3

Calculate the O.S. of all the atoms in the following species :

(i) $ClO^{\ominus}$, (ii) $NO_2^{\ominus}$, (iii) $NO_3^{\ominus}$, (iv) CCl_4

(v) K_2CrO_4 and (vi) $KMnO_4$

Solution:

(i) In $ClO^{\ominus}$, the net charge on the species is –1 and therefore the sum of the oxidation states of Cl and O must be equal to –1. Oxygen will have an O.S. of –2 and if the O.S. of Cl is assumed to be 'x' then x – 2 should be equal to –1.

x is + 1

(ii) $NO_2^{\ominus}$: $(2 \times -2) + x = -1$ (where 'x' is O.S. of N)

$\therefore$ x = +3

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(iii) $NO_3^{\ominus}$: $x + (3 \times -2) = -1$ (where 'x' is O.S. of N)

$x = +5$

(iv) In CCl_4, Cl has an O.S. of –1

$x + 4 \times -1 = 0$

$\therefore\ x = +4$ (where 'x' is O.S. of C)

(v) K_2CrO_4 : K has O.S. of +1 and O has O.S. of –2 and let Cr has O.S. 'x' then, $(2 \times +1) + x + (4 \times -2) = 0$

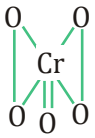
$\therefore\ x = +6$

(vi) $KMnO_4$: $+1 + x + (4 \times -2) = 0$

$\therefore\ x = +7$ (where x is O.S. of Mn).

Miscellaneous examples:

In order to determine the exact or individual oxidation number we need to take help from the structures of the molecules. Some special cases are discussed as follows :

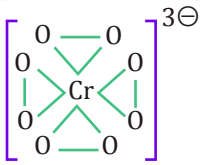
- The structure of CrO_5 is 

  From the structure it is evident that in CrO_5 there are two peroxide linkages and one double bond. The contribution of each peroxide linkage is –2. Let the O.N. of Cr is x.

  $\therefore\ x + (-2)2 + (-2) = 0$ or $x = 6$

  $\therefore$ O.N. of Cr = +6 **Ans.**

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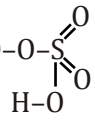
K_3CrO_8 

- From the structure it is evident that in $CrO_8^{3\ominus}$ there are four peroxide linkages.

  The contribution of each peroxide linkage is –2. Let the O.N. of Cr is x.

  $\therefore\ x + (-2)4 = -3$ or $x = +5$

  $\therefore$ O.N. of Cr = +5 **Ans.**

- The structure of H_2SO_5 is 

  From the structure, it is evident that in H_2SO_5. there is one peroxide linkage, two sulphur - oxygen double bond and one OH group. Let the O.N. of S = x.

  $\therefore\ +1 - 2 + x + (-2)\ 2 + (-2) + 1 = 0$

  or $x + 2 - 8 = 0$

  or $x - 6 = 0$

  or $x = 6$

  $\therefore$ O.N. of S in H_2SO_5 is +6 **Ans.**

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Paradox of fractional oxidation number:

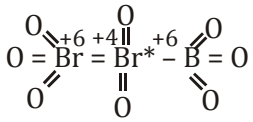
Fractional oxidation state is the average oxidation state of the element under examination and the structural parameters reveal that the element for whom fractional oxidation state is realised is actually present in different oxidation states. Structure of the species C_3O_2, Br_3O_8 and $S_4O_6^{2\ominus}$ reveal the following bonding situations.

- The element marked with asterisk in each species is exhibiting the different oxidation state (oxidation number) from rest of the atoms of the same element in each of the species. This reveals that in C_3O_2, two carbon atoms are present in +2 oxidation state each whereas the third one is present in zero oxidation state and the average is 4/3. However, the realistic picture is +2 for two terminal carbons and zero for the middle carbon.

$$O=\overset{+2}{C}=\overset{0}{C}^{*}=\overset{+2}{C}=O$$

Structure of C_3O_2
(Carbon suboxide)

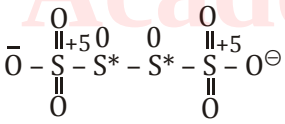
- Likewise, in Br_3O_8, each of the two terminal bromine atoms are present in +6 oxidation state and the middle bromine is present in +4 oxidation state. Once again the average, that is different from reality is 16/3.



Structure of Br_3O_8 (tribromooctaoxide)

- In the same fashion, in the species $S_4O_6^{2\ominus}$, is 2.5, whereas the reality being +5,0,0 and +5 oxidation number respectively for each sulphur.

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Structure of $S_4O_6^{2\ominus}$ (tetrathionate ion)

Fractional Oxidation states:

Most of elements show fractional oxidation states. For example, oxidation state of oxygen in superoxide's of alkali metals (KO_2, SO_2, RbO_2) is –1/2.

Example : In Fe_3O_4, Fe shows its oxidation state as 8/3 as it is a mixed oxide and can be written as $FeO.Fe_2O_3$. Similarly, oxidation states of Boron in B_4O_{10} and B_5H_9 is 2.5 and 1.8 respectively.

Oxidising and Reducing agent:

Oxidising agent or Oxidant:

Oxidising agents are those compounds which can oxidise others and reduces itself during the chemical reaction. Those reagents whose O.N. decrease or which gain electrons in a redox reaction are termed as oxidants

e.g. $KMnO_4$, $K_2Cr_2O_7$, HNO_3, conc. H_2SO_4 etc, are powerful oxidising agents.

Reducing agent or Reductant:

Reducing agents are those compounds which can reduce others and oxidise itself during the chemical reaction. Those reagents whose O.N. increase or which loses electrons in a redox reaction are termed as reductants.

e.g. KI, $Na_2S_2O_3$ are powerful reducing agents.

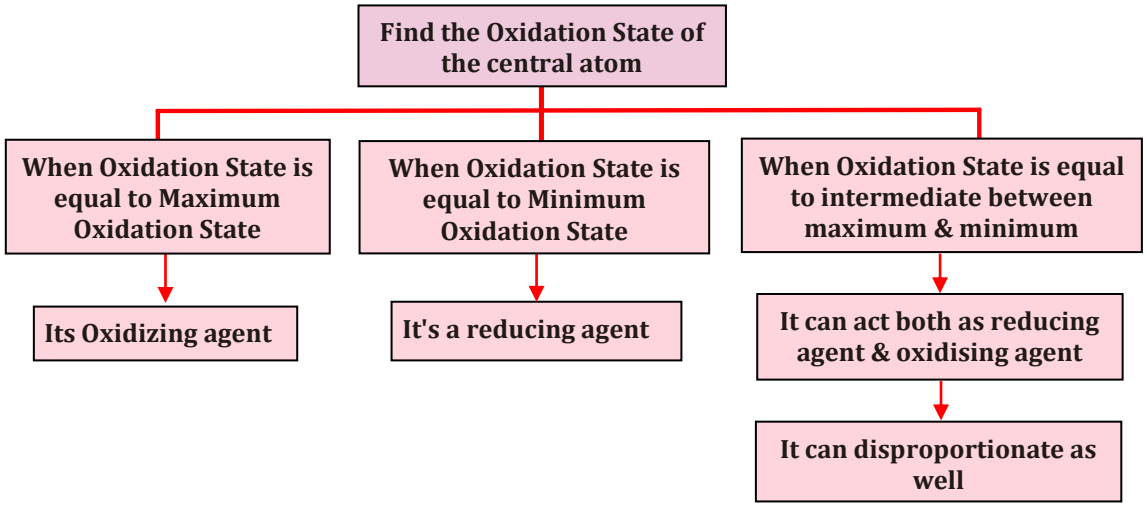
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Note :

There are some compounds also which can work both oxidising agent and reducing agent.

e.g. H_2O_2, $NO_2^{\ominus}$

How to identify whether a particular substance is an oxidising or reducing agent:



Ex. Identify the species undergoing oxidation and reduction

(i) $\overset{(O)}{\overbrace{\underset{(R)}{\underbrace{N_2 + H_2 \longrightarrow NH_3}}}}$

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(ii) $\overset{(R)}{\overbrace{\underset{(O)}{\underbrace{Zn + HCl \longrightarrow ZnCl_2 + H_2}}}}$

(iii) $\overset{(R)}{\overbrace{\underset{(O)}{\underbrace{Mg + O_2 \longrightarrow MgO}}}}$

(iv) $\overset{(O)}{\overbrace{\underset{(R)}{\underbrace{Zn + NaOH \longrightarrow Na_2ZnO_2 + H_2}}}}$

(v) $\overset{(R)}{\overbrace{\underset{(O)}{\underbrace{H_2SO_4 + Mg \longrightarrow MgSO_4 + SO_2 + H_2}}}}$

(vi) $H_2SO_4 + NaOH \longrightarrow Na_2SO_4 + H_2O$

Neither O & R

(vii) $\overset{(R)}{\overbrace{\underset{(O)}{\underbrace{Cl_2 + NaOH \longrightarrow NaCl + NaClO_3 + H_2O}}}}$

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(viii) $\overset{(R)}{\overbrace{FeCl_3 + SnCl_2}} \longrightarrow \underbrace{FeCl_2 + SnCl_4}_{(O)}$

(ix) $AgNO_3 + NaCl \longrightarrow AgCl + NaNO_3$

No oxidation and no reduction.

Valency concept of oxidation and reduction:

(i) Oxidation : According to this concept, it is the process in which increase in (+) ve valency or decrease in (–) ve valency of a substance takes place.

Example :

$$\underset{(0)}{2Mg} + O_2 \longrightarrow \underset{(2)}{2MgO}$$

Oxidation

(ii) Reduction : According to this concept, it is the process in which (+) ve valency decreases whereas (–) ve valency increases.

Example :

$$\underset{(+2)}{2HgCl_2} + \underset{(+2)}{SnCl_2} \longrightarrow \underset{(+1)}{Hg_2Cl_2} + \underset{(+4)}{SnCl_4}$$

Reduction

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Oxidation number concept:

(i) Oxidation: According to this concept, the oxidation number of an element in a reaction increases.

(ii) Reduction: According to this concept, the oxidation number of an element in a reaction is decrease.

Example :

$$6H^{\oplus} + \underset{\quad Mn^{7\oplus}}{2MnO_4^{\ominus}} + \underset{S^{2\ominus}}{5H_2S} \longrightarrow \underset{Mn^{2\oplus}}{2Mn^{2\oplus}} + \underset{S^0}{5S} + 8H_2O$$

Reduction

Oxidation

Special Note:

Redox reaction involves two half reactions, one involving loss of electron called half oxidation reaction and the other involving gain of electron called half reduction reaction.

Example :

$$2Hg^{2\oplus} + Sn^{2\oplus} \longrightarrow Hg_2^{1\oplus} + Sn^{4\oplus}$$

$-2e^{\ominus}$ Oxidation

$+2e^{\oplus}$ Reduction

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Redox Reactions:

(a) The reactions in which oxidation and reduction both occur simultaneously are called redox reactions.

(b) Most of the chemical reactions are redox because if one element is there to lose electrons, other element has to be there to accept them.

(c) Any redox reaction may be divided in two parts.

(i) Oxidation half reaction.

(ii) Reduction half reaction.

Now, we will study some reaction.

S. No.	Reaction	Oxidant (Getting Reduced)	Reductant (Getting Oxidised)
1.	$C + O_2 \longrightarrow CO_2$	O [0 ⟶ –2]	C [0 ⟶ +4]
2.	$PbS + 4H_2O_2 \longrightarrow PbSO_4 + 4H_2O$	O [–1 ⟶ –2]	S [–2 ⟶ +6]
3.	$Sn + 2F_2 \longrightarrow SnF_4$	F [0 ⟶ –1]	Sn [0 ⟶ +4]
4.	$SO_2 + 2H_2O + Cl_2 \longrightarrow 2HCl + H_2SO_4$	Cl [0 ⟶ –1]	S [+4 ⟶ +6]
5.	$I_2 + 10HNO_3 \longrightarrow 2HIO_3 + 10NO_2 + 4H_2O$	N [+5 ⟶ +4]	I [0 ⟶ +5]
6.	$CuO + H_2 \longrightarrow Cu + H_2O$	Cu [+2 ⟶ 0]	H [0 ⟶ +1]
7.	$2KMnO_4 + 3H_2SO_4 + 5H_2S \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5S$	Mn [+7 ⟶ +2]	S [–2 ⟶ 0]
8.	$H_2O_2 + Ag_2O \longrightarrow 2Ag + H_2O + O_2$ (Oxygen of H_2O_2)	Ag [+1 ⟶ 0]	O [–1 ⟶ 0]
9.	$H_2SO_4 + 2HI \longrightarrow SO_2 + I_2 + 2H_2O$	S [+6 ⟶ +4]	I [–1 ⟶ 0]

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Note : In reaction 2 oxygens of ozone have different oxidation state.

Structure of Ozone is $O_3 \leftarrow O_1 = O_2$

oxidation state of $O_1 = +2$

oxidation state of $O_2 = 0$

oxidation state of $O_3 = -2$

Type of reaction:

(a) Combination Reactions :

When two or more elements combine together to form compound then such reactions are called redox reactions.

For example,

$A + B \longrightarrow C$

(a) $3\overset{0}{Mg}(s) + \overset{0}{N_2}(g) \xrightarrow{\Delta} \overset{+2}{Mg_3}\overset{-3}{N_2}(s)$

(b) $\overset{0}{C}(s) + \overset{0}{O_2}(g) \xrightarrow{\Delta} \overset{+4\ -2}{CO_2}(g)$

(c) $\overset{-4\ +1}{CH_4}(g) + 2\overset{0}{O_2}(g) \xrightarrow{\Delta} \overset{+4\ -2}{CO_2}(g) + 2\overset{+1\ -2}{H_2O}(l)$

(d) $\overset{0}{Li}(s) + \overset{0}{N_2}(g) \xrightarrow{\Delta} \overset{+1\ -3}{Li_3N}(s)$

In reaction (c), there is no change in the oxidation number of hydrogens.

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(b) Decomposition Reactions:

Decomposition reactions are opposite of combination reactions. Decomposition reactions are those in which when a molecule breaks down to form two or more components, at least one of them must be in the elemental state. For example :

(a) $\overset{+1}{2K}\overset{+5}{Cl}\overset{-2}{O_3}(s) \xrightarrow{\Delta} \overset{+1}{2K}\overset{-1}{Cl}(s) + 3\overset{0}{O_2}(g)$

(b) $\overset{+2}{Ca}\overset{-1}{H_2} \xrightarrow{\Delta} \overset{0}{Ca}(s) + \overset{0}{H_2}(g)$

(c) $\overset{+1}{2Na}\overset{-1}{H}(s) \xrightarrow{\Delta} 2\overset{0}{Na}(s) + \overset{0}{H_2}(g)$

(d) $\overset{+1}{H_2}\overset{-2}{O}(\ell) \xrightarrow{\Delta} 2\overset{0}{H_2}(g) + \overset{0}{O_2}(g)$

Exception :

$\overset{+2}{Ca}\overset{+4}{C}\overset{-2}{O_3}(s) \xrightarrow{\Delta} \overset{+2}{Ca}\overset{-2}{O}(s) + \overset{+4}{C}\overset{-2}{O_2}(g)$

(c) Displacement Reactions:

A reaction in which an atom or anion in a compound is replaced by an atom or on ion of another element is called a displacement reaction. It may be denoted as :

$X + YZ \longrightarrow XZ + Y$

(a) Metal displacement reactions

(b) Non-metal displacement reactions

(i) Metal displacement reaction :- In these reactions, a metal in the compound is displaced by another metal in the uncombined state.

For example :

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$\overset{+2}{Cu}\overset{+6}{S}\overset{-2}{O_4}(aq) + \overset{0}{Zn}(s) \xrightarrow{\Delta} \overset{0}{Cu}(s) + \overset{+2}{Zn}\overset{+6}{S}\overset{-2}{O_4}(aq)$

$\overset{+3}{Cr_2}\overset{-2}{O_3}(s) + 2\overset{0}{Al}(s) \xrightarrow{\Delta} 2\overset{0}{Cr}(s) + \overset{+3}{Al_2}\overset{-2}{O_3}(s)$

$\overset{+5}{V_2}\overset{-2}{O_5}(s) + 2\overset{0}{Ca}(s) \xrightarrow{\Delta} 5\overset{+2}{Ca}\overset{-2}{O}(s) + 2\overset{0}{V}(s)$

$\overset{+4}{Ti}\overset{-1}{Cl_4}(\ell) + 2\overset{0}{Mg}(s) \xrightarrow{\Delta} Ti(s) + 2MgCl_2(s)$

(ii) Non-metal displacement reaction :- In these reactions, a metal or non-metal displaces another non-metal from its compound. These non-metal displacement redox reactions include hydrogen displacement and a rarely occurring reaction involving oxygen displacement.

For example :

$\overset{0}{2K}(s) + 2\overset{+1}{H}\overset{-2}{O}(\ell) \xrightarrow{\Delta} 2\overset{+1}{K}\overset{-2}{O}\overset{+1}{H}(aq) + \overset{0}{H_2}(g)$

$\overset{0}{Ba}(s) + 2\overset{+1}{H_2}\overset{-2}{O}(\ell) \xrightarrow{\Delta} 2\overset{+2}{Ba}(\overset{-2}{O}\overset{+1}{H})_2(aq) + \overset{0}{H_2}(g)$

$\overset{0}{Mg}(s) + 2\overset{+1}{H_2}\overset{-2}{O}(\ell) \xrightarrow{\Delta} \overset{+2}{Mg}(\overset{-2}{O}\overset{+1}{H})_2(s) + \overset{0}{H_2}(g)$

$\overset{0}{2Fe}(s) + 3\overset{+1}{H_2}\overset{-2}{O}(\ell) \xrightarrow{\Delta} \overset{+3}{Fe_2}\overset{-2}{O_3}(s) + 3\overset{0}{H_2}(g)$

$\overset{0}{Zn}(s) + 2\overset{+1}{H}\overset{-1}{Cl}(aq) \xrightarrow{\Delta} \overset{+2}{Zn}\overset{-1}{Cl_2}(aq) + \overset{0}{H_2}(g)$

$\overset{0}{Fe}(s) + 2\overset{+1}{H}\overset{-1}{Cl}(aq) \xrightarrow{\Delta} \overset{+2}{Fe}\overset{-1}{Cl_2}(aq) + \overset{0}{H_2}(g)$

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(d) Disproportionation reactions :

A redox reaction in which a same element present in a particular compound in a definite oxidation state is oxidized as well as reduced simultaneously.

Disproportionation reactions are a special type of redox reactions. One of the reactants in a disproportionation reaction always contains **an element that can exist in atleast three oxidation states.** The element in the form of reacting substance is in the intermediate oxidation state and both higher and lower oxidation states of that element are formed in the reaction.

$\overset{+1\ -1}{2H_2O_2}\ (aq) \longrightarrow H_2O\ (\ell) + O_2(g)$

$\overset{0}{S_8}(s) + 12OH^{\ominus}(aq) \longrightarrow 4\overset{-2}{S^{2\ominus}}\ (aq) + 2\overset{+2}{S_2}O_3^{2\ominus}\ (aq) + 6H_2O\ (\ell)$

$\overset{0}{Cl_2}\ (g) + 2OH^{\ominus}\ (aq) \longrightarrow \overset{+1}{ClO^{\ominus}}\ (aq) + \overset{-1}{Cl^{\ominus}}\ (aq) + H_2O\ (\ell)$

Consider following reactions:

(a) $2KClO_3 = 2KCl + 3O_2$

$KClO_3$ plays a role of oxidant and reductant both. Because same element is not oxidised and reduced. Here, Cl present in $KClO_3$ is reduced and O present in $KClO_3$ is oxidized. So, **it's not a disproportion reaction** although it looks like one.

(b) $4KClO_3 \longrightarrow 3KClO_4 + KCl$

It's a case of disproportionation reaction in which Cl is the atom disproportionating.

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List of some important disproportionation reaction:

1. $H_2O_2 \longrightarrow H_2O + O_2$
2. $X_2 + OH^{\ominus}\ (dil.) \longrightarrow X^{\ominus} + XO^{\ominus}$
3. $X_2 + OH^{\ominus}\ (conc.) \longrightarrow X^{\ominus} + XO_3^{\ominus}$

F_2 does not (cannot) undergo disproportionation as it is the most electronegative element.

$F_2 + NaOH\ (dil) \longrightarrow F^{\ominus} + OF_2$

$F_2 + NaOH$ concentration $(dil) \longrightarrow F^{\ominus} + O_2$

Reverse of disproportionation is called comproportionation. In some of the disproportionation reactions by changing the medium (from acidic to basic or reverse) the reaction goes in backward direction and can be taken as an example of comproportionation reactions.

$I^{\ominus} + IO_3^{\ominus} + H^{\oplus} \longrightarrow I_2 + H_2O$ (acidic)

(e) Comproportionation reaction:

$NH_4NO_2 \longrightarrow N_2 + 2H_2O$

Nitrogen in this compound has –3 and +3 oxidation number so it is not a definite value, so it's not a disproportion reaction. It's an example of **comproportionating reaction** which is a class of redox reaction in which an element from two different oxidation state gets converted into a single oxidation state.

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Equivalent weight (E):

$$\text{Eq. wt (E)} = \frac{\text{Molecular weight}}{\text{valency factor (v.f)}} = \frac{\text{Mol.wt.}}{\text{n-factor}}$$

$$\text{no of Equivalents} = \frac{\text{mass of a sample}}{\text{eq.wt. of that species}}$$

- Equivalent mass is a pure number when expressed in gram, it is called gram equivalent mass.
- The equivalent mass of substance may have different values under different conditions.

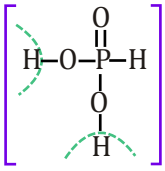
Valency factor calculation :

For Acids :

valence factor = number of replaceable $H^{\oplus}$ ions

Illustration 4

HCl, H_2SO_4 H_3PO_4 H_3PO_3



{see there are only two replaceable $H^{\oplus}$ ions}

Solution:

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valence → 1 2 3 2

factor (assume 100% dissociation)

Eq. wt. → $\frac{M}{1}$ $\frac{M}{2}$ $\frac{M}{3}$ $\frac{M}{2}$

Illustration 5

Find the valence factor for following acids

(i) CH_3COOH (ii) NaH_2PO_4 (iii) H_3BO_3

Solution:

(i) 1 (ii) 2 (iii) 1

For Base :

v.f. = number of replicable $OH^{\ominus}$ ions

Illustration 6

NaOH KOH

Solution:

v.f. → 1 1

E. → $\frac{M}{1}$ $\frac{M}{1}$

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Illustration 7

Find the valence factor for following bases

(i) $Ca(OH)_2$ (ii) $CsOH$ (iii) $Al(OH)_3$

Solution:

(i) 2 (ii) 1 (iii) 3

Acid - base reaction:

In case of acid base reaction, the valence factor is the actual number of $H^{\oplus}$ or $OH^{\ominus}$ replaced in the reaction.

The acid or base may contain more number of replaceable $H^{\oplus}$ or $OH^{\ominus}$ than actually replaced in reaction.

v.f. for base is the number of $H^{\oplus}$ ion from the acid replaced by per molecule of the base.

Illustration 8

$2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$

Base Acid

Solution:

valency factor of base = 1

valency factor of acid = 2

Here two molecule of NaOH replaced $2H^{\oplus}$ ion from the H_2SO_4 therefore per molecule of NaOH replaced only one $H^{\oplus}$ ion of acid so v.f. = 1

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v.f. for acid is number of $OH^{\ominus}$ replaced for the base by per molecule of acid

Illustration 9

$NaOH + H_3PO_4 \rightarrow NaH_2PO_4 + H_2O$

Base Acid

Solution:

valence factor of acid = 1

here one of molecule of H_3PO_4 replaced one $OH^{\ominus}$ from NaOH therefore v.f. for H_3PO_4 is = 1

$$E = \frac{\text{mol.wt.of } H_3PO_4}{1}$$

Illustration 10

$2NaOH + H_3PO_4 \rightarrow Na_2HPO_4 + 2H_2O$

Base Acid

Solution:

valence factor of acid = 2

here one of molecule of H_3PO_4 replaced two $OH^{\ominus}$ from NaOH therefore v.f. for H_3PO_4 is = 2

$$E = \frac{\text{mol.wt.of } H_3PO_4}{2}$$

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Illustration 11

$Al(OH)_3 + HCl \rightarrow Al(OH)_2Cl + H_2O$

Base Acid

Solution:

valence factor of base = 1

here one of molecule of $Al(OH)_3$ replaced one $H^{\oplus}$ from HCl therefore v.f. for $Al(OH)_3$ is = 1

$E = \frac{\text{mol.wt.of } Al(OH)_3}{1}$

For Salts :

v.f. = Total number of positive charge or negative charge present into the compound.

Illustration 12

Find the eq. weight for the following

Na_2CO_3 $Fe_2(SO_4)_3$ $FeSO_4.7H_2O$

Solution:

	Na_2CO_3	$Fe_2(SO_4)_3$	$FeSO_4.7H_2O$
v.f.	2	$2 \times 3 = 6$	2
E.	$\frac{M}{2}$	$\frac{M}{6}$	$\frac{M}{2}$

Eq. wt. of oxidising / reducing agents in redox reaction:

The equivalent weight of an oxidising agent is that weight which accepts one mole electron in a chemical reaction.

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(a) Equivalent wt. of an oxidant (get reduced)

$= \frac{\text{Mol.wt.}}{\text{No. of electrons gained by one mole}}$

Ex. In acidic medium

$6e^{\ominus} + Cr_2O_7^{2\ominus} + 14H^{\oplus} \longrightarrow 2Cr^{3\oplus} + 7H_2O$

Eq. wt. of $K_2Cr_2O_7 = \frac{\text{Mol. wt of } K_2Cr_2O_7}{6} = \frac{\text{Mol. wt.}}{6}$

Note :

[6 in denominator indicates that 6 electrons were gained by $Cr_2O_7^{2\ominus}$ as it is clear from the given balanced equation]

(b) Similarly, equivalent wt. of a reductant (gets oxidised)

$= \frac{\text{Mol. wt.}}{\text{No. of electrons lost by one mole}}$

Ex. In acidic medium,

$C_2O_4^{2\ominus} \longrightarrow 2CO_2 + 2e^{\ominus}$

Here, total electrons lost = 2

So, eq. wt. = $\frac{\text{Mol. wt.}}{2}$

(c) In different condition a compound may have different equivalent weights. Because, it depends upon the number of electrons gained or lost by that compound in that reaction.

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Ex. (i) $MnO_4^{\ominus} \longrightarrow Mn^{2\oplus}$ (acidic medium)
(+7) (+2)

Here 5 electrons are taken so eq. wt. = $\frac{\text{Mol.wt.of } KMnO_4}{5} = \frac{158}{5} = 31.6$

(ii) $MnO_4^{\ominus} \longrightarrow Mn^{4\oplus}$ (neutral medium)
(+7) (+4)

Here, only 3 electrons are gained, so eq. wt. = $\frac{\text{Mol.wt.of } KMnO_4}{3} = \frac{158}{3} = 52.7$

(iii) $MnO_4^{\ominus} \longrightarrow MnO_4^{2\ominus}$ (alkaline medium)
(+7) (+6)

Here, only one electron is gained, so eq. wt. = $\frac{\text{Mol.wt.of } KMnO_4}{1} = 158$

Note :

It is important to note that $KMnO_4$ acts as an oxidant in every medium although with different strength which follows the order.

acidic medium > neutral medium > alkaline medium

Ex. $2S_2O_3^{2\ominus} \longrightarrow S_4O_6^{2\ominus} + 2e^{\ominus}$
(Reducing agent)

equivalent weight of $S_2O_3^{\ominus} = \frac{2M}{2} = 1M$

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Questions based on Equivalent weight :

Illustration 13

Molecular weight of $KMnO_4$ in acidic medium and neutral medium will be respectively:

(A) 7 × equivalent wt. and 2 × equivalent wt. (B) 5 × equivalent wt. and 3 × equivalent wt.

(C) 4 × equivalent wt. and 5 × equivalent wt. (D) 2 × equivalent wt. and 4 × equivalent wt.

Solution:

In Acidic Medium

$Mn^{7\oplus} \longrightarrow Mn^{2\oplus}$

5 × eq. w = molecular weight

In neutral medium → $Mn^{7\oplus} \longrightarrow Mn^{4\oplus}$

3 × equivalent wt. = molecular weight

Illustration 14

In acidic medium, equivalent weight of $K_2Cr_2O_7$ (Mol. wt. = M) is –

(A) M/3 (B) M/4 (C) M/6 (D) M/2

Solution:

$K_2Cr_2O_7 \rightarrow Cr^{3\oplus}$

v.f = 6

Eq. wt. = $\frac{M}{6}$

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For Acid-Base (Neutralization Reaction) Or Redox Reaction:

$N_1V_1 = N_2V_2$ is always true.

But $M_1V_1 = M_2V_2$ (may or may not be true)

But $M_1 \times n_1 \times V_1 = M_2 \times n_2 \times V_2$ (always true where n terms represent n-factor).

Some oxidizing agents/reducing agents with Eq. wt.

Species	Changed to	Reaction exchanged or change in O.N.	Electrons	Eq. wt.
$MnO_4^{\ominus}$ (O.A.)	$Mn^{2\oplus}$ in acidic medium	$MnO_4^{\ominus} + 8H^{\oplus} + 5e^{\ominus} \longrightarrow Mn^{2\oplus} + 4H_2O$	5	$E = \frac{M}{5}$
$MnO_4^{\ominus}$ (O.A.)	MnO_2 in neutral medium	$MnO_4^{\ominus} + 3e^{\ominus} + 2H_2O \longrightarrow MnO_2 + 4OH^{\ominus}$	3	$E = \frac{M}{3}$
$MnO_4^{\ominus}$ (O.A.)	$MnO_4^{2\ominus}$ in basic medium	$MnO_4^{\ominus} + e^{\ominus} \longrightarrow MnO_4^{2\ominus}$	1	$E = \frac{M}{1}$
$Cr_2O_7^{2\ominus}$(O.A.)	$Cr^{3\oplus}$ in acidic medium	$CrO_7^{2\ominus} + 14H^{\oplus} + 6e^{\ominus} \longrightarrow 2Cr^{3\oplus} + 7H_2O$	6	$E = \frac{M}{6}$
MnO_2(O.A.)	$Mn^{2\oplus}$ in acidic medium	$MnO_2 + 4H^{\oplus} + 2e^{\ominus} \longrightarrow Mn^{2\oplus} + 2H_2O$	2	$E = \frac{M}{2}$
Cl_2(O.A.) in bleaching powder	$Cl^{\ominus}$	$Cl_2 + 2e^{\ominus} \longrightarrow 2Cl^{\ominus}$	2	$E = \frac{M}{2}$
$CuSO_4$(O.A.) in iodometric titration	$Cu^{\oplus}$	$Cu^{2\oplus} + e^{\ominus} \longrightarrow Cu^{\oplus}$	1	$E = \frac{M}{1}$
$S_2O_3^{2\ominus}$ (R.A.)	$S_4O_6^{2\ominus}$	$2S_2O_3^{2\ominus} \longrightarrow S_4O_6^{2\ominus} + 2e^{\ominus}$ (for two molecules)	2	$E = \frac{M}{2}$
H_2O_2(O.A.)	H_2O	$H_2O_2 + 2H^{\oplus} + 2e^{\ominus} \longrightarrow 2H_2O$	2	$E = \frac{M}{2}$
H_2O_2(R.A.)	O_2	$H_2O_2 \longrightarrow O_2 + 2H^{\oplus} + 2e^{\ominus}$ (O.N. of oxygen in H_2O_2 is –1 per atom)	2	$E = \frac{M}{2}$
$Fe^{2\oplus}$(R.A.)	$Fe^{3\oplus}$	$Fe^{2\oplus} \longrightarrow Fe^{3\oplus} + e^{\ominus}$	1	$E = \frac{M}{1}$

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Illustration 15

Find the n-factor in the following chemical changes.

(i) $KMnO_4 \xrightarrow{H^{\oplus}} Mn^{2\oplus}$ (ii) $KMnO_4 \xrightarrow{H_2O} Mn^{4\oplus}$ (iii) $KMnO_4 \xrightarrow{OH^{\ominus}} Mn^{6\oplus}$

(iv) $K_2Cr_2O_7 \xrightarrow{H^{\oplus}} Cr^{3\oplus}$ (v) $C_2O_4^{2\ominus} \rightarrow CO_2$ (vi) $FeSO_4 \rightarrow Fe_2O_3$

(vii) $Fe_2O_3 \rightarrow FeSO_4$

Solution:

(i) In this reaction, $KMnO_4$ which is an oxidizing agent, itself gets reduced to $Mn^{2\oplus}$ under acidic conditions.
$n = |1 \times (+7) - 1 \times (+2)| = 5$

(ii) In this reaction, $KMnO_4$ gets reduced to $Mn^{4\oplus}$ under neutral or slightly (weakly) basic conditions.
$n = |1 \times (+7) - 1 \times (+4)| = 3$

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(iii) In this reaction, $KMnO_4$ gets reduced to $Mn^{6\oplus}$ under basic conditions.
$n = |1 \times (+7) - 1 \times (+6)| = 1$

(iv) In this reaction, $K_2Cr_2O_7$ which acts as an oxidizing agent reduced to $Cr^{3\oplus}$ under acidic conditions. (It does not react under basic conditions.)
$n = |2 \times (+6) - 2 \times (+3)| = 6$

(v) In this reaction, $C_2O_4^{2\ominus}$ (oxalate ion) gets oxidized to CO_2 when it is reacted with an oxidizing agent.
$n = |2 \times (+3) - 2 \times (+4)| = 2$

(vi) In this reaction, ferrous ions get oxidized to ferric ions.
$n = |1 \times (+2) - 1 \times (+3)| = 1$

(vi) In this reaction, ferric ions are getting reduced to ferrous ions.
$n = |2 \times (+3) - 2 \times (+2)| = 2$

Illustration 16

Calculate the molar ratio in which the following two substances would react ?
$Ba_3(PO_4)_2$ and $AlCl_3$

Solution:

n-factor of $Ba_3(PO_4)_2 = 3 \times (+2) = 6 = n_1$
While n-factor of $AlCl_3 = 1 \times (+3) = 3 = n_2$

$\frac{n_1}{n_2} = \frac{6}{3}$, If $\frac{n_1}{n_2} = \frac{x}{y}$

Molar ratio = $\frac{y}{x}$ (inverse of equivalent ratio)

$\therefore$ Molar ratio in which $Ba_3(PO_4)_2$ and $AlCl_3$ will react = 3 : 6 = 1 : 2

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Balancing of Redox Reaction:

All balanced equations must satisfy two criteria

1. **Atom balance (mass balance) :**
   That is there should be the same number of atoms of each kind in reactant and products side.
2. **Charge balance :**
   That is the sum of actual charges on both side of the equation must be equal.
   There are two methods for balancing the redox equations.
   (a) Oxidation - number change method
   (b) Ion electron method or half cell method

   (a) Oxidation number change method :

   This method was given by Jonson. In a balanced redox reaction, total increase in oxidation number must be equal to total decreases in oxidation number. This equivalence provides the basis for balancing redox reactions.

   The general procedure involves the following steps :

   (i) Select the atom in oxidising agent whose oxidation number decreases and indicate the gain of electrons.

   (ii) Select the atom in reducing agent whose oxidation number increases and write the loss of electrons.

   (iii) Now cross multiply i.e. multiply oxidising agent by the number of loss of electrons and reducing agent by number of gain of electrons.

   (iv) Balance the number of atoms on both sides whose oxidation numbers change in the reaction.

   (v) In order to balance oxygen atoms, add H_2O molecules to the side deficient in oxygen. Then balance the number of H atoms by adding $H^{\oplus}$ ions in the hydrogen.

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Illustration 17

Write the skeleton equation for each of the following processes and balance them by ion electron method :

(i) Permanganate ion oxidizes oxalate ions in acidic medium to carbon dioxide and gets reduced itself to $Mn^{2\oplus}$ ions.

(ii) Chlorine reacts with base to form chlorate ion, chloride ion and water in acidic medium.

Solution:

(i) The skeleton equation for the process :

$MnO_4^{\ominus} + C_2O_4^{2\ominus} + H^{\oplus} \longrightarrow Mn^{2\oplus} + CO_2 + H_2O$

Step (1) : Indicating oxidation number :

$\overset{+7}{Mn}\overset{-2}{O_4^{\ominus}} + \overset{+3}{C_2}\overset{2\ominus}{O_4} \longrightarrow Mn^{2\oplus} + \overset{+4}{C}\overset{-2}{O_2} + \overset{+1}{H_2}\overset{-2}{O}$

Step (2) : Writing oxidation and reduction half reaction :

$\overset{+3}{C_2}\overset{2\ominus}{O_4} \longrightarrow 2\overset{+4}{C}O_2$ (Oxidation half)

$\overset{+7}{Mn}O_4^{\ominus} \longrightarrow Mn^{2\oplus}$ (Reduction half)

Step (3) : Adding electrons to make the difference in O.N.

$\overset{+3}{C_2}\overset{2\ominus}{O_4} \longrightarrow 2\overset{+4}{C}O_2 + 2e^{\ominus}$

$\overset{+7}{Mn}O_4^{\ominus} + 5e^{\ominus} \longrightarrow Mn^{2\oplus}$

Step (4) : Balancing 'O' atom by adding H_2O molecules

$C_2O_4^{2\ominus} \longrightarrow 2CO_2 + 2e^{\ominus}$

$MnO_4^{\ominus} + 5e^{\ominus} \longrightarrow Mn^{2\oplus} + 4H_2O$

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Step (5) : Balancing H atom by adding $H^{\oplus}$ ions

$C_2O_4^{2\ominus} \longrightarrow 2CO_2 + 2e^{\ominus}$

$MnO_4^{\ominus} + 5e^{\ominus} + 8H^{\oplus} \longrightarrow Mn^{2\oplus} + 4H_2O$

Step (6) : Multiply the oxidation half reaction by 2 and reduction half reaction by 5 to equalize the electrons lost and gained and add the two half reactions.

$[C_2O_4^{2\ominus} \longrightarrow 2CO_2 + 2e^{\ominus}] \times 5$

$[MnO_4^{\ominus} + 5e^{\ominus} + 8H^{\oplus} \longrightarrow Mn^{2\oplus} + 4H_2O] \times 2$

$2MnO_4^{\ominus} + 5C_2O_4^{2\ominus} + 16H^{\oplus} \longrightarrow 10CO_2 + 2Mn^{2\oplus} + 8H_2O$

(ii) The skeleton equation for the given process :

$Cl_2 + OH^{\ominus} \longrightarrow Cl^{\ominus} + ClO_3^{\ominus} + H_2O$

Step (1) : Indicate the oxidation number of each atom

$\overset{0}{Cl_2} + \overset{2\ominus}{O}H^{\ominus} \longrightarrow Cl^{\ominus} + \overset{+5}{Cl}O_3^{\ominus} + \overset{+1}{H_2}\overset{-2}{O}$

Thus, chlorine is the only element which undergoes the change in oxidation number. It decreases its oxidation number from 0 to 1 and also increases its oxidation number from 0 to 5.

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Step (2) : Write the oxidation and reduction half reactions

$\overset{0}{Cl_2} \longrightarrow 2\overset{+5}{ClO_3^{\ominus}}$ (Oxidation half)

$\overset{0}{Cl_2} \longrightarrow 2Cl^{\ominus}$ (Reduction half)

Step (3) : Add electrons to make up for the difference in O.N.

$\overset{0}{Cl_2} \longrightarrow 2\overset{+5}{ClO_3^{\ominus}} + 10e^{\ominus}$

$\overset{0}{Cl_2} + 2e^{\ominus} \longrightarrow 2Cl^{\ominus}$

Step (4) : Balance O atoms by adding H_2O molecules

$Cl_2 + 6H_2O \longrightarrow 2(ClO_3)^{\ominus} + 10e^{\ominus}$

$Cl_2 + 2e^{\ominus} \longrightarrow Cl^{\ominus}$

Step (5) : Since medium is basic, balance H atoms by adding H_2O molecules to the side falling short of H atoms and equal number of $OH^{\ominus}$ ions to the other side.

$Cl_2 + 6H_2O + 12OH^{\ominus} \longrightarrow 2ClO_3^{\ominus} + 10e^{\ominus} + 12H_2O$

$Cl_2 + 2e^{\ominus} \longrightarrow 2Cl^{\ominus}$

Step (6) : Multiply the reduction half reaction by 5 and add two half reactions.

$Cl_2 + 5H_2O + 2OH^{\ominus} \longrightarrow 2ClO_3^{\ominus} + 10e^{\ominus} + 12H_2O$

$[Cl_2 + 2e^{\ominus} \longrightarrow 2Cl^{\ominus}] \times 5$

$Cl_2 + 5Cl_2 + 12OH^{\ominus} \longrightarrow 2ClO_3^{\ominus} + 10Cl^{\ominus} + 6H_2O$

or, $6Cl_2 + 12OH^{\ominus} \longrightarrow 2ClO_3^{\ominus} + 10Cl^{\ominus} + 6H_2O$

or, $3Cl_2 + 6OH^{\ominus} \longrightarrow ClO_3^{\ominus} + 5Cl^{\ominus} + 3H_2O$

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Illustration 18

Balance the following chemical reaction by oxidation number method and write their skeleton equation :

(i) The nitrate ions in acidic medium oxidize magnesium to $Mg^{2\oplus}$ ions but itself gets reduced to nitrous oxide.

Solution:

(i) The skeleton equation for the given process is

$Mg + NO_3^{\ominus} \longrightarrow Mg^{2\oplus} + N_2O + H_2O$

Step (1) : $\overset{0}{Mg} + \overset{+5\ -2}{(NO_3)^{\ominus}} \longrightarrow Mg^{2\oplus} + \overset{+1\ -2}{N_2O} + \overset{+1\ -2}{H_2O}$

Multiply $NO_3^{\ominus}$ by 2 to equalize N atoms

Step (2) :

O.N. increases by 2 per Mg atom

$\overset{0}{Mg} + 2NO_3^{\ominus} \longrightarrow Mg^{2\oplus} + \overset{+1}{N_2O} + H_2O$

$4 \times 2 = 8$

O.N. decreases by 4 per N atom

Step (3) : Equalize increase/decrease in O.N. by multiplying Mg by 4 and $2NO_3^{\ominus}$ by 1.

$4Mg + 2NO_3^{\ominus} \longrightarrow Mg^{2\oplus} + N_2O + H_2O$

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Step (4) : Balance atoms other than O and H

$4Mg + 2NO_3^{\ominus} \longrightarrow 4Mg^{2\oplus} + N_2O + H_2O$

Step (5) : Balance O atoms

$4Mg + 2NO_3^{\ominus} \longrightarrow 4Mg^{2\oplus} + N_2O + H_2O + 4H_2O$

Step (6) : Balance H atoms as is done in acidic medium.

$4Mg + 2NO_3^{\ominus} + 10H^{\oplus} \longrightarrow 4Mg^{2\oplus} + N_2O + 5H_2O$

(b) Ion electron method or half cell method :

By this method redox equation are balanced in two different medium

(a) Acidic medium (b) Basic medium

Balancing in acidic medium:

Students are advised to follow the following steps to balance the redox reactions by ion electron method in acidic medium.

Illustration 19

Balance the following redox reaction.

$FeSO_4 + KMnO_4 + H_2SO_4 \longrightarrow Fe_2(SO_4)_3 + MnSO_4 + H_2O$

Solution:

Step-I assign the oxidation No. to each elements present in the reaction.

$\overset{+2\ +6\ -2}{Fe\,S\,O_4} + \overset{+1\ +7\ -2}{K\,Mn\,O_4} + \overset{+1\ +6\ -2}{H_2\,S\,O_4} \longrightarrow \overset{+3\ +6\ -2}{Fe_2(S\,O_4)_3} + \overset{+2\ +6\ -2}{Mn\,S\,O_4} + \overset{+1\ -2}{H_2\,O}$

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Step-II Now convert the reaction in ionic form by eliminating the elements or species which are not going either oxidation or reduction

$Fe^{2\oplus} + \overset{+7}{Mn}O_4^{\ominus} \longrightarrow Fe^{3\oplus} + Mn^{2\oplus}$

Step-III Now identify the oxidation / reduction occurring into the reaction.

Undergoes reduction

$Fe^{2\oplus} + MnO_4^{\ominus} \longrightarrow Fe^{3\oplus} + Mn^{2\oplus}$

Undergoes oxidation

Step-IV Split the ionic reaction in two half one for oxidation and other for reduction

$Fe^{2\oplus} \xrightarrow{\text{oxidation}} Fe^{3\oplus}$ $MnO_4^{\ominus} \xrightarrow{\text{Reduction}} Mn^{2\oplus}$

Step-V Balance the atom other than oxygen and hydrogen atom in both half reactions

$Fe^{2\oplus} \rightarrow Fe^{3\oplus}$ $MnO_4^{\ominus} \rightarrow Mn^{2\oplus}$

Fe & Mn atom are balanced in both side.

Step-VI Now balance O & H atom by H_2O & $H^{\oplus}$ respectively by the following way for one excess oxygen atom add one H_2O on the other side and two $H^{\oplus}$ on the same side.

$Fe^{2\oplus} \longrightarrow Fe^{3\oplus}$ (no oxygen atom) ...(i)

$8H^{\oplus} + MnO_4^{\ominus} \longrightarrow Mn^{2\oplus} + 4H_2O$ ...(ii)

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Step VII Now see equation (i) & (ii) is balanced atom wise. Now balance both equations charge wise to balance the charge add electrons to the electrically positive side.

$Fe^{2\oplus} \xrightarrow{\text{oxidation}} Fe^{3\oplus} + e^{\ominus}$ ...(1)

$5e^{\ominus} + 8H^{\oplus} + MnO_4^{\ominus} \xrightarrow{\text{Reduction}} Mn^{2\oplus} + 4H_2O$ ...(2)

Step VIII The number of electrons gained and lost in each half-reaction are equalised by multiplying suitable factor in both the half reaction and finally the half reactions are added to give the overall balanced reaction.

Here we multiply equation (i) by 5 and (ii) by one

$Fe^{2\oplus} \longrightarrow Fe^{3\oplus} + e^{\ominus}$ ...(1) × 5

$5e^{\ominus} + 8H^{\oplus} + MnO_4^{\ominus} \rightarrow M^{2\oplus} + 4H_2O$ ...(2) × 1

$5Fe^{2\oplus} + 8H^{\oplus} + MnO_4^{\ominus} \rightarrow 5Fe^{3\oplus} + Mn^{2\oplus} + 4H_2O$

(Here at this stage you will get balanced redox reaction in ionic form)

Step IX Now convert the ionic reaction into molecular form by adding the elements or species which are removed in step (II).

Now by some manipulation you will get

$5\ FeSO_4 + KMnO_4 + 4H_2SO_4 \rightarrow \frac{5}{2}\ Fe_2(SO_4)_3 + MnSO_4 + 4H_2O$ or

$10\ FeSO_4 + 2KMnO_4 + 8H_2SO_4 \rightarrow 5Fe_2(SO_4)_3 + 2MnSO_4 + 8H_2O + K_2SO_4$

Illustration 20

Balance the following redox reaction ?

$Cr(OH)_3 + IO_3^{\ominus} \xrightarrow{OH^{\ominus}} I^{\ominus} + CrO_4^{2\ominus}$

Solution:

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(a) $Cr(OH)_3 + IO_3^{\ominus} \xrightarrow{OH^{\ominus}} I^{\ominus} + CrO_4^{2\ominus}$

(b) Separate the two half reactions.

$Cr(OH)_3 \longrightarrow CrO_4^{2\ominus}$ (Oxidation half reaction)

$IO_3^{\ominus} \longrightarrow I^{\ominus}$ (Reduction half reaction)

(c) Balance O-atoms by adding H_2O.

$H_2O + Cr(OH)_3 \longrightarrow CrO_4^{2\ominus}$

$IO_3^{\ominus} \longrightarrow I^{\ominus} + 3H_2O$

(d) Balance H-atoms by adding H_2O to side having deficiency and an equal number of $OH^{\ominus}$ ions to the other side ($\therefore$ medium is known)

$5OH^{\ominus} + H_2O + Cr(OH)_3 \longrightarrow CrO_4^{2\ominus} + 5H_2O$

or $5OH^{\ominus} + Cr(OH)_3 \longrightarrow CrO_4^{2\ominus} + 4H_2O$

$IO_3^{\ominus} + 6H_2O \longrightarrow I^{\ominus} + 3H_2O + 6OH^{\ominus}$

or $IO_3^{\ominus} + 3H_2O \longrightarrow I^{\ominus} + 6OH^{\ominus}$

(e) Balance the charges by electrons

$5OH^{\ominus} + Cr(OH)_3 \longrightarrow CrO_4^{2\ominus} + 4H_2O + 3e^{\ominus}$

$IO_3^{\ominus} + 6H_2O + 6e^{\ominus} \longrightarrow I^{\ominus} + 3H_2O + 6OH^{\ominus}$

(f) Multiply first equation by 2 and add to second to give

$10OH^{\ominus} + 2Cr(OH)_3 \longrightarrow 2CrO_4^{2\ominus} + 8H_2O + 6e^{\ominus}$

$IO_3^{\ominus} + 6H_2O + 6e^{\ominus} \longrightarrow I^{\ominus} + 3H_2O + 6OH^{\ominus}$

$4OH^{\ominus} + 2Cr(OH)_3 + IO_3^{\ominus} \longrightarrow 5H_2O + 2CrO_4^{2\ominus} + I^{\ominus}$

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Law of equivalence:

The law states that one equivalent of an element combine with one equivalent of the other, and in a chemical reaction equivalent and milli equivalent of reactants react in equal amount to give same no. of equivalent or milli equivalents of products separately.

According :

(i) $aA + bB \rightarrow mM + nN$

m.eq of A = m.eq of B = m.eq of M = m.eq of N

(ii) In a compound M_xN_y

m.eq of M_xN_y = m.eq of M = m.eq of N

Illustration 21

The number of moles of oxalate ions oxidized by one mole of $MnO_4^{\ominus}$ ion in acidic medium.

(A) $\frac{5}{2}$ (B) $\frac{2}{5}$ (C) $\frac{3}{5}$ (D) $\frac{5}{3}$

Solution:

Equivalents of $C_2O_4^{2\ominus}$ = equivalents of $MnO_4^{\ominus}$

x (mole) × 2 = 1 × 5

$x = \frac{5}{2}$

Drawbacks of Equivalent concept :

(i) Since equivalent weight of a substance for example oxidising or reducing agent may be variable hence it is better to use mole concept.

e.g. $5e^{\ominus} + 8H^{\oplus} + MnO_4^{\ominus} \rightarrow Mn^{2\oplus} + 2H_2O$

$\therefore$ Eq. wt of $MnO_4^{\ominus} = \frac{MnO_4^{\ominus}(\text{mol.wt.})}{5}$

e.g. $3e^{\ominus} + 2H_2O + MnO_4^{\ominus} \rightarrow MnO_2 + 4OH^{\ominus}$

$\therefore$ Eq. wt. of $MnO_4^{\ominus} = \frac{MnO_4^{\ominus}}{3}$

Thus the no. of equivalents of $MnO_4^{\ominus}$ will be different in the above two cases but no. of moles will be same.

(ii) Normality of any solution depends on reaction while molarity does not.

For example.

Consider 0.1 mol $KMnO_4$ dissolved in water to make 1L solution. Molarity of this solution is 0.1 M. However, its normality is not fixed. It will depend upon the reaction in which $KMnO_4$ participates, e.g. If $KMnO_4$ forms $Mn^{2\oplus}$, normality = 0.1 × 5 = 0.5 N. This same sample of $KMnO_4$, if employed in a reaction giving MnO_2 as product (Mn in +4 state) will have normality 0.1 × 3 = 0.3 N.

(iii) The concept of equivalents is handy, but it should be used with care. One must never equate equivalents in a sequence which involves same element in more than two oxidation states. Consider an example, KIO_3 reacts with KI to liberate iodine and liberated iodine is titrated with standard hypo solution, the reactions are:

(i) $IO_3^{\ominus} + I^{\ominus} \longrightarrow I_2$

(ii) $I_2 + S_2O_3^{2\ominus} \longrightarrow S_4O_6^{2\ominus} + I^{\ominus}$

meq of hypo = meq of I_2 = meq of $IO_3^{\ominus}$ + meq of $I^{\ominus}$

$\because$ $IO_3^{\ominus}$ react with $I^{\ominus} \Rightarrow$ meq of $IO_3^{\ominus}$ = meq of $I^{\ominus}$

meq of hypo = 2 × meq of $IO_3^{\ominus}$

This is wrong. Note that I_2 formed by (i) have v.f. = 5/3 & reacted in equation (ii) have v.f. = 2.

$\because$ v.f. of I_2 in both the equation are different therefore we cannot equate m.eq is sequence. In this type of case students are advised to use mole concept.

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Illustration 22

How many millilitres of 0.02000 M $KMnO_4$ solution would be required to exactly titrate 25.00 mL of 0.2000 M $Fe(NO_3)_2$ solution.

Solution:

Method - 1 : Mole concept method

Starting with 25.00 mL of 0.2000M$Fe^{2\oplus}$, we can write.

Millimoles of $Fe^{2\oplus}$ = 25.00 × 0.2000

and in volume V (in millilitres of the $MnO_4^{\ominus}$)

Millimoles of $MnO_4^{\ominus}$ = V (0.02000)

The balanced reaction is :

$MnO_4^{\ominus} + 5Fe^{2\oplus} + 8H^{\oplus} \longrightarrow Mn^{2\oplus} + 5\ Fe^{3\oplus} + 4H_2O$

This requires that at the equivalent point,

$$\frac{V(0.02000)}{1} = \frac{(25.00)(0.2000)}{5} \quad V = 50.00 \text{ mL}$$

Method - 2 : Equivalent Method :

Equivalents of $MnO_4^{\ominus}$ = 5 × moles of $MnO_4^{\ominus}$

Normality of $MnO_4^{\ominus}$ = 5 × molarity of $MnO_4^{\ominus}$

For $Fe^{2\oplus}$, moles and equivalents are equal,

At the equivalence point,

Equivalents of $MnO_4^{\ominus}$ = Equivalents of $Fe^{2\oplus}$

or $V_{MnO_4^{\ominus}}$ × Normality of $MnO_4^{\ominus}$

= $V_{Fe^{2\oplus}}$ × Normality of $Fe^{2\oplus}$

For 0.02000 M $MnO_4^{\ominus}$ solution

Normality of $MnO_4^{\ominus}$= (5) (0.02000) = 0.1 N and for 0.2000 M $Fe^{2\oplus}$ solution

Normality of $Fe^{2\oplus}$ = 0.2000 N, $V_{MnO_4^{\ominus}}$ = (25.00 mL) $\left(\frac{0.2000}{0.1000}\right)$ = 50.00 mL

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Applications of the law of equivalence:

Simple Titration:

In this, we can find the concentration of unknown solution by reacting it with solution of known concentration (standard solution).

For example, let there be a solution of substance A of unknown concentration. We are given solution of another substance B whose concentration is known (N_1). We take a certain known volume of A in a flask (V_2) and then we add B to A slowly till all the A is consumed by B (this can be known with the help of indicators). Let us, assume that the volume of B consumed is V_1. According to the law of equivalence, the number of g equivalents of B at the end point.

$\therefore N_1V_1 = N_2V_2$, where N_2 is the conc. of A.

From this we can calculate the value of N_2.

Illustration 23

1.20 g sample of Na_2CO_3 and K_2CO_3 was dissolved in water to form 100 mL of a Solution. 20 mL of this solution required 40 mL of 0.1 N HCl for complete neutralization. Calculate the weight of Na_2CO_3 in the mixture. If another 20 mL of this solution is treated with excess of $BaCl_2$ what will be the weight of the precipitate ?

Solution:

Let, weight of Na_2CO_3 = x g

Weight of K_2CO_3 = y g

$\therefore$ x + y = 1.20 g ...(i)

For neutralization reaction of 100 mL

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Meq. of Na_2CO_3 + Meq. of K_2CO_3 = Meq. of HCl

$\Rightarrow \frac{x}{106} \times 2 \times 1000 + \frac{y}{138} \times 2 \times 1000 = \frac{40 \times 0.1 \times 100}{20}$

$\therefore$ 69 x + 53 y = 73.14 ...(ii)

From Eqs. (i) and (ii), we get

x = 0.5962 g

y = 0.604 g

Solution of Na_2CO_3 and K_2CO_3 gives ppt. of $BaCO_3$ with $BaCl_2$

(Meq. of Na_2CO_3 + Meq. of K_2CO_3) in 20 mL = Meq. of $BaCO_3$

$\Rightarrow$ Meq. of HCl for 20 mL mixture = Meq. of $BaCO_3$

$\Rightarrow$ Meq. of $BaCO_3 = 40 \times 0.1 = 4$

$\frac{W_{BaCO_3}}{E_{BaCO_3}} \times 1000 = 40 \times 0.1 = 4$

$\frac{W_{BaCO_3}}{197} \times 2 \times 1000 = 4$

$\therefore \quad W_{BaCO_3} = 0.394$ g

Iodometric and iodimetric titration:

The reduction of free iodine to iodide ions and oxidation of iodide ions to free iodine occurs in these titrations.

$I_2 + 2e^{\ominus} \longrightarrow 2I^{\ominus}$ (reduction)

$2I^{\ominus} \longrightarrow I_2 + 2e^{\ominus}$ (oxidation)

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These are divided into two types :

(i) Iodometric Titration :

In iodometric titrations, an oxidizing agent is allowed to react in neutral medium or in acidic medium with excess of potassium iodide to liberate free iodine.

KI + oxidizing agent $\longrightarrow$ I_2

Free iodine is titrated against a standard reducing agent usually with sodium thiosulphate. Halogen, dichromates, cupric ion, peroxides etc., can be estimated by this method.

$I_2 + Na_2S_2O_3 \longrightarrow 2NaI + Na_2S_4O_6$

$2CuSO_4 + 4KI \longrightarrow Cu_2I_2 + 2K_2SO_4 + I_2$

$K_2Cr_2O_7 + 6KI + 7H_2SO_4 \longrightarrow Cr_2(SO_4)_3 + 4K_2SO_4 + 7H_2O + 3I_2$

(ii) Iodimetric Titration

These are the titrations in which free iodine is used as it is difficult to prepare the solution of iodine (volatile and less soluble in water), it is dissolved in KI solution :

$KI + I_2 \longrightarrow KI_3$ (Potassium triiodide)

This solution is first standardized before using with the standard solution of substance such as sulphite, thiosulphate, arsenite etc, are estimated.

In iodimetric and iodometric titration, starch solution is used as an indicator. Starch solution gives blue or violet colour with free iodine. At the end point, the blue or violet colour disappears when iodine is completely changed to iodide.

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Some iodometric titrations (titrating solutions is $Na_2S_2O_3.5H_2O$):

Estimation of	Reaction	Relation between O.A. and R.A.
I_2	$I_2 + 2Na_2S_2O_3 \longrightarrow 2NaI + Na_2S_4O_6$ Or $I_2 + 2S_2O_3^{2\ominus} \longrightarrow 2I^{\ominus} + S_4O_6^{2\ominus}$	$I_2 \equiv 2I \equiv 2Na_2S_2O_3$ Eq. wt. of $Na_2S_2O_3$ = M/1
$CuSO_4$	$2CuSO_4 + 4KI \longrightarrow 2Cu_2I_2 + 2K_2SO_4 + I_2$ $Cu^{2\oplus} + 4I^{\ominus} \longrightarrow Cu_2I_2 + I_2$ (White ppt.) $CaOCl_2 + H_2O \longrightarrow Ca(OH)_2 + Cl_2$	$2CuSO_4 \equiv I_2 \equiv 2I \equiv 2Na_2S_2O_3$ Eq. wt. of $CuSO_4$ = M/1
$CaOCl_2$	$Cl_2 + 2KI \longrightarrow 2KCl + I_2$ Or $Cl_2 + 2I^{\ominus} \longrightarrow 2Cl^{\ominus} + I_2$	$CaOCl_2 \equiv Cl_2 \equiv I_2 \equiv 2I \equiv 2Na_2S_2O_3$ Eq. wt. of $CaOCl_2$ = M/2
MnO_2	$MnO_2+4HCl(conc) \xrightarrow{\Delta} MnCl_2+Cl_2+2H_2O$ $Cl_2 + 2KI \longrightarrow 2KCl + I_2$ or $MnO_2 + 4H^{\oplus}+2Cl^{\ominus} \longrightarrow Mn^{2\oplus} +2H_2O+Cl_2$ $Cl_2 + 2I^{\ominus} \longrightarrow I_2 + 2Cl^{\ominus}$	$MnO_2 \equiv Cl_2 \equiv I_2 \equiv 2I \equiv 2Na_2S_2O_3$ Eq. wt. of MnO_2 = M/2
$IO_3^{\ominus}$	$IO_3^{\ominus} + 5I^{\ominus} + 6H^{\oplus} \longrightarrow 3I_2 + 3H_2O$	$IO_3^{\ominus} \equiv 3I_2 \equiv 6I \equiv 6Na_2S_2O_3$ Eq. wt. of $IO_3^{\ominus}$ = M/6
H_2O_2	$H_2O_2 + 2I^{\ominus} + 2H^{\oplus} \longrightarrow I_2 + 2H_2O$	$H_2O_2 \equiv I_2 \equiv 2I \equiv 2Na_2S_2O_3$ Eq. wt. of H_2O_2 = M/2
Cl_2	$Cl_2 + 2I^{\ominus} \longrightarrow 2Cl^{\ominus} + I_2$	$Cl_2 \equiv I_2 \equiv 2I \equiv 2Na_2S_2O_3$ Eq. wt. of Cl_2 = M/2
O_3	$O_3 + 6I^{\ominus} + 6H^{\oplus} \longrightarrow 3I_2 + 3H_2O$	$O_3 \equiv 3I_2 \equiv 6I \equiv 6Na_2S_2O_3$ Eq. wt. of O_3 = M/6
$Cr_2O_7^{2\ominus}$	$Cr_2O_7^{2\ominus}+14H^{\oplus} + 6I^{\ominus} \longrightarrow 3I_2 + 2Cr^{3\oplus}+7H_2O$	$Cr_2O_7^{2\ominus} \equiv 3I_2 \equiv 6I \equiv 6Na_2S_2O_3$ Eq. wt. of $Cr_2O_7^{2\ominus}$ = M/6
$MnO_4^{\ominus}$	$2MnO_4^{\ominus}+10I^{\ominus}+16H^{\oplus} \longrightarrow 2MnO_4^{\ominus} + 5I_2 + 8H_2O$	$2MnO_4^{\ominus} \equiv 5I_2 \equiv 10I \equiv 10Na_2S_2O_3$ Eq. wt. of $MnO_4^{\ominus}$ = M/5
$BrO_3^{\ominus}$	$BrO_3^{\ominus}+6I^{\ominus}+6H^{\oplus} \longrightarrow Br^{\ominus} + 3I_2 + 3H_2O$	$BrO_3^{\ominus} \equiv 3I_2 \equiv 6I \equiv 6Na_2S_2O_3$ Eq. wt. of $BrO_3^{\ominus}$ = M/6
As(V)	$H_3AsO_4 + 2I^{\ominus} + 2H^{\oplus} \longrightarrow H_3AsO_3 + H_2O + I_2$	$H_3AsO_4 \equiv I_2 \equiv 2I \equiv 2Na_2S_2O_3$ Eq. wt. of H_3AsO_4 = M/2

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Some iodimetric titration (titrating solutions is I_2 in KI):

ESTIMATION OF	REACTION	RELATION BETWEEN O.A. AND R.A.
H_2S (IN ACIDIC MEDIUM)	$H_2S + I_2 \longrightarrow S + 2I^{\ominus} + 2H^{\oplus}$	$H_2S \equiv I_2 \equiv 2I$ EQ. WT. OF $H_2S = M/2$
$SO_3^{2\ominus}$ (IN ACIDIC MEDIUM)	$SO_3^{2\ominus} + I_2 + H_2O \longrightarrow SO_3^{2\ominus} + 2I^{\ominus} + 2H^{\oplus}$	$SO_3^{2\ominus} \equiv I_2 \equiv 2I$ EQ. WT. OF $SO_3^{2\ominus} = M/2$
$Sn^{2\oplus}$ (IN ACIDIC MEDIUM)	$Sn^{2\oplus} + I_2 \longrightarrow Sn^{4\oplus} + 2I^{\ominus}$	$Sn^{2\oplus} \equiv I_2 \equiv 2I$ EQ. WT. OF $Sn^{2\oplus} = M/2$
As(III) (AT pH = 8)	$H_2AsO_3^{\ominus} + I_2 + H_2O \longrightarrow HAsO_4^{2\ominus} + 2I^{\ominus} + 2H^{\oplus}$	$H_2AsO_3^{\ominus} \equiv I_2 \equiv 2I$ EQ. WT. OF $H_2AsO_3^{\ominus} = M/2$
N_2H_4	$N_2H_4 + 2I_2 \longrightarrow N_2 + 4H^{\oplus} + 4I^{\ominus}$	$N_2H_4 \equiv 2I_2 \equiv 4I$ EQ. WT. OF $N_2H_4 = M/4$

Back titration:

Back titration is used to calculate % purity of a sample. Let us assume that we are given an impure solid substance C weighing 'w' gm and we are asked to calculate the percentage of pure C in the sample. We will assume that the impurities are inert. We are provided with two solutions A and B, where the concentration of B is known (N_1) and that of A is not known. This type of titration will work only if the following condition is satisfied, i.e. the nature of compounds A, B and C should be such that A and B can react with each other. A and C can react with each other but the product of A and C should not react with B.

Now, we take a certain volume of A in a flask (the g equivalents of A taken should be ≥ g equivalents of C in the sample and this can be done by taking A in excess). Now, we perform a simple titration using B. Let us assume that the volume of B used is V_1. In another beaker, we again take the solution of A in the same volume as taken earlier. Now, C is added to this and after the reaction is completed, the solution is being titrated with B. Let us assume that the volume of B used up is V_2. Gram equivalents of B used in the first titration = N_1V_1.

∴ gm. equivalents of A initially = N_1V_1

gm. equivalents of B used in the second titration = N_1V_2

∴ gm. equivalents of A left in excess after reacting with C = N_1V_2

gm. equivalents of A that reacted with C = $N_1V_1 - N_1V_2$

If the n-factor of C is x, then the moles of pure C = $\frac{N_1V_1 - N_1V_2}{x}$

∴ The weight of C = $\frac{N_1V_1 - N_1V_2}{x} \times$ Molecular weight of C

∴ Percentage of C = $\frac{N_1V_1 - N_1V_2}{x} \times \frac{\text{Molecular wt. of C}}{w} \times 100$

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Some redox titrations (excluding iodometric/iodimetric):

Estimation of	By titrating with	Reactions	Relation between O.A. and R.A.
$Fe^{2\oplus}$	$MnO_4^{\ominus}$	$Fe^{2\oplus} \longrightarrow Fe^{3\oplus} + e^{\ominus}$ $MnO_4^{\ominus} + 8H^{\oplus} + 5e^{\ominus} \longrightarrow Mn^{2\oplus} + 4H_2O$	$5Fe^{2\oplus} \equiv MnO_4^{\ominus}$ Eq. wt. $Fe^{2\oplus}$ = M/1 Eq. wt. $MnO_4^{\ominus}$= M/5
$Fe^{2\oplus}$	$Cr_2O_7^{2\ominus}$	$Fe^{2\oplus} \longrightarrow Fe^{3\oplus} + e^{\ominus}$ $Cr_2O_7^{2\ominus} + 14H^{\oplus} + 6e^{\ominus} \longrightarrow 2Cr^{3\oplus} + 7H_2O$	$6Fe^{2\oplus} \equiv Cr_2O_7^{2\ominus}$ Eq. wt. $Cr_2O_7^{2\ominus}$ = M/6
$C_2O_4^{2\ominus}$	$MnO_4^{\ominus}$	$C_2O_4^{2\ominus} \longrightarrow 2CO_2 + 2e^{\ominus}$ $MnO_4^{\ominus} + 8H^{\oplus} + 5e^{\ominus} \longrightarrow Mn^{\oplus 2} + 4H_2O$	$5C_2O_4^{2\ominus} \equiv 2MnO_4^{\ominus}$ Eq. wt. $C_2O_4^{2\ominus}$ = M/2 Eq. wt. $MnO_4^{\ominus}$= M/5
H_2O_2	$MnO_4^{\ominus}$	$H_2O_2 \longrightarrow 2H^{\oplus} + O_2 + 2e^{\ominus}$ $MnO_4^{\ominus} + 8H^{\oplus} + 5e^{\ominus} \longrightarrow Mn^{2\oplus} + 4H_2O$	$5H_2O_2 \equiv 2MnO_4^{\ominus}$ Eq. wt. H_2O_2 = M/2 Eq. wt. $MnO_4^{\ominus}$= M/5
As_2O_3	$MnO_4^{\ominus}$	$As_2O_3 + 5H_2O \rightarrow 2AsO_4^{3\ominus} + 10H^{\oplus} + 4e^{\ominus}$	Eq. wt. As_2O_3= M/4
$AsO_3^{3\ominus}$	$BrO_3^{\ominus}$	$AsO_3^{3\ominus} + H_2O \longrightarrow AsO_4^{3\ominus} + 2H^{\oplus} + 2e^{\ominus}$ $BrO_3^{\ominus} + 6H^{\oplus} + 6e^{\ominus} \longrightarrow Br^{\ominus} + 3H_2O$	Eq. wt. $AsO_3^{3\ominus}$= M/2 Eq. wt. $BrO_3^{\ominus}$ = M/6

Hardness of water:

Hard water does not give lather with soap. Water free from soluble salts of calcium and magnesium is called **Soft water**. It gives lather with soap easily.

Hard water forms scum/precipitate with soap. Soap containing sodium stearate ($C_{17}H_{35}COONa$) reacts with hard water to precipitate out Ca/Mg stearate.

$$2C_{17}H_{35}COONa(aq) + M^{2\oplus}(aq) \rightarrow (C_{17}H_{35}COO)_2M \downarrow + 2Na^{\oplus}(aq);\ M \text{ is Ca/Mg}$$

Hardness of water is of two types

(a) Temporary Hardness

(b) Permanent Hardness

(a) Temporary Hardness :-

This is due to presence of bicarbonate of calcium and magnesium. Rain water dissolves a small quantities of CO_2 from the atmosphere forming a very dilute solution of carbonic acid.

This water attacks Ca and Mg carbonate in any rock

$$CaCO_3 + H_2O + CO_2 \longrightarrow Ca(HCO_3)_2$$

Temporary hardness in water is easily removed by boiling, as the bicarbonates decomposes readily and the insoluble carbonates are precipitated.

$$Ca(HCO_3)_2 \xrightarrow{\text{Boil}} CaCO_3 + H_2O + CO_2$$

(Insoluble)

$$Mg(HCO_3)_2 \xrightarrow{\text{Boil}} MgCO_3 + H_2O + CO_2$$

(Insoluble)

Temporary hardness can also be removed by Clark's process which involves the addition of slaked lime $Ca(OH)_2$.

$$Ca(HCO_3)_2 + Ca(OH)_2 \xrightarrow{\text{Boil}} 2CaCO_3 + 2H_2O$$

(Insoluble)

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(b) Permanent Hardness :- Permanent hardness is due to presence of sulphates and chlorides of both calcium and magnesium. This type of hardness cannot be removed by boiling or by $Ca(OH)_2$;. Substance used to remove the hardness of water are known as water softener. This various water softeners are –

(i) Washing soda :- It removes both the temporary and permanent hardness by converting soluble Ca and Mg compounds into insoluble Carbonates.

$CaCl_2 + Na_2CO_3 \longrightarrow CaCO_3 + 2NaCl$

$CaSO_4 + Na_2CO_3 \longrightarrow CaCO_3 + Na_2SO_4$

$Ca(HCO_3)_2 + Na_2CO_3 \longrightarrow CaCO_3 + NaHCO_3$

(Insoluble) (Insoluble)

(ii) Permutit : Permutit is technical name given to certain hydrated silicates of aluminium and sodium. or sodium alumunium orthosilicate or sodium zeolite.

Example : $Na_2Al_2Si_2O_8.H_2O$

$Na_2Al_2Si_2O_8.H_2O + Ca^{2\oplus} \longrightarrow CaAl_2Si_2O_8.H_2O + 2Na^{\oplus}$

$Na_2Al_2Si_2O_8.H_2O + Mg^{2\oplus} \longrightarrow MgAl_2Si_2O_8.H_2O + 2Na^{\oplus}$

These ions can be re-exchanged by treating it with brine (NaCl) solution.

$CaAl_2Si_2O_8.H_2O + 2NaCl \longrightarrow Na_2Al_2Si_2O_8.H_2O + CaCl_2$

This method is useful for the removal of both temporary and permanent hardness of water.

(iii) CALGON: The complex salt of metaphosphoric acid, sodium hexametaphosphate $[Na(PO_3)]_6$ is called Calgon. It is represented as $Na_2[Na_4(PO_3)_6]$.

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$2CaSO_4 + Na_2[Na_4(PO_3)_6] \longrightarrow Na_2[Ca_2(PO_3)_6] + Na_2SO_4$

$2MgSO_4 + Na_2[Na_4(PO_3)_6] \longrightarrow Na_2[Mg_2(PO_3)_6] + Na_2SO_4$

(iv) Ion exchange Resins:- Ion exchange resins are the most popular water softener these days. These resins are synthetic substances. The cation exchanger consists of granular insoluble organic acid resins having giant molecules with $-SO_3H$ or -COOH groups while the anion exchange contains giant organic molecules with basic groups derived from amine. Ion exchange resins removes all soluble mineral from water.

Hard water $\longrightarrow$ [ ] $\longrightarrow$ [ ] $\longrightarrow$ Pure water

Cation exchanger	Anion exchanger
Remove cation like	Remove anion like
$Na^{\oplus}$, $Mg^{2\oplus}$ or $Ca^{2\oplus}$	$SO_4^{2\ominus}$, $Cl^{\ominus}$ or $NO_3^{\oplus}$
and furnish $H^{\oplus}$	and furnish $OH^{\ominus}$
$H^{\oplus} + OH^{\ominus} \longrightarrow$	H_2O
$Ca^{2\oplus} + RH_2 \longrightarrow$	$R_2Ca + 2H^{\oplus}$

The water coming from cation exchanger is acidic due to $H^{\oplus}$ ions. This water is then passed through another bed containing anion exchanger. This exchanger removes anion like $Cl^{\ominus}$, $SO_4^{2\ominus}$, NO_3 by exchanging with $OH^{\ominus}$ ions.

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Reaction at Cation exchanger :

$Ca^{2\oplus} + RH_2 \longrightarrow R_2Ca + 2H^{\oplus}$

$Mg^{2\oplus} + RH_2 \longrightarrow R_2Mg + 2H^{\oplus}$

Reaction at Anion exchanger :

$R_x(OH)_2 + Cl^{\ominus} \longrightarrow RCl_2 + OH^{\ominus}$

$R_x(OH)_2 + SO_4^{2\ominus} \longrightarrow RSO_4 + OH^{\ominus}$

Note :

Degree of Hardness of water is measured in terms of ppm of $CaCO_3$ which is defined as number of parts of $CaCO_3$ by mass present in one million parts by mass of water whose equivalent is equal to the gram equivalent of various calcium & magnesium salts.

For **example** consider the sample of hard water which is found to contain 36 mg of $MgSO_4$ per kg of water.

gm equivalent of $CaCO_3$ = gm eq. of $MgSO_4$

$$\frac{W_{CaCO_3}}{100}\times 2 = \frac{36\times 10^{3\ominus}}{120}\times 2$$

$$W_{CaCO_3} = 3 \times 10^{2\ominus}\text{gm}$$

$$\text{degree of hardness} = \text{ppm of } CaCO_3 = \frac{3\times 10^{2\ominus}}{1000}\times 10^6 = 30$$

Illustration 24

A 3.00 g sample containing Fe_3O_4, Fe_2O_3 and an inert impure substance, is treated with excess of KI solution in presence of dilute H_2SO_4. The entire iron is converted into $Fe^{2\oplus}$ along with the liberation of iodine. The resulting solution is diluted to 100 mL. A 20 mL of the diluted solution require 11 mL of 0.5 M $Na_2S_2O_3$ solution to reduce the iodine present. A 50 mL of diluted solution after complete extraction of the iodine requires 12.80 mL of 0.25 $KMnO_4$ solution in dilute H_2SO_4 medium for the oxidation of $Fe^{2\oplus}$. Calculate the percentages of Fe_2O_3 and Fe_3O_4 in the original sample.

Solution:

Let number of moles of Fe_3O_4 is 'a' and no. of moles of Fe_2O_3 is 'b' in 3 gram sample.

Now, 'a' moles Fe_3O_4 means (a moles Fe_2O_3 + a mole of FeO)

So, in sample (a + b) moles Fe_2O_3 and 'a' mole FeO present.

$Fe_2O_3 + KI + H_2SO_4 \longrightarrow FeSO_4 + I_2 + K_2SO_4 + H_2O$

No. of gram equivalent of Fe_2O_3 = no. of gram equivalent of I_2

$$\text{mole of } Fe_2O_3 \times n_f(Fe_2O_3) = \text{mole of } I_2 \times n_f(I_2)$$

$(a + b) \times 2 = n_{I_2} \times 2$

n_{I_2} = (a + b) moles.

Now, solution is diluted to 100 ml

$$\text{So, } [I_2] = \frac{n}{V_{litre}} = \frac{(a+b)}{100/1000} = 10(a+b)$$

$$[Fe^{2\oplus}] = \left(\frac{n(Fe^{2\oplus})}{V_{litre}}\right) = \frac{(3a+2b)}{100/1000} = 10(3a + 2b)$$

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Now 20 ml of this solution titrated with 11 ml, 0.5 M $Na_2S_2O_3$

$I_2 + Na_2S_2O_3 \rightarrow NaI + Na_2S_4O_6$

No. of gram equivalent of I_2 = no. of gram equivalent of $Na_2S_2O_3$

$(N \times V)_{I_2} = (N \times V)_{Na_2S_2O_3}$

$(M \times n_f \times V)_{I_2} = (M \times n_f \times V)_{Na_2S_2O_3}$

$10\,(a + b) \times 2 \times 20 = 0.5 \times 1 \times 11$

$$a + b = \frac{11}{800} \qquad ...(i)$$

50 ml of solution after extraction of I_2 allowed for oxidation

of $Fe^{2\oplus}$ by 12.80 ml, 0.25 M $KMnO_4$.

$Fe^{2\oplus} + KMnO_4 + H_2SO_4 \rightarrow Fe^{3\oplus} + MnSO_4 + K_2SO_4 + H_2O$

no. of gram equivalent of $Fe^{2\oplus}$ = no. of gram equivalent of $KMnO_4$.

$(M \times n_f \times V)_{Fe^{2\oplus}} = (M \times n_f \times V)_{KMnO_4}$

$10(3a + 2b) \times 1 \times 50 = 0.25 \times 5 \times 12.80$

$$3a + 2b = \frac{16}{500} \qquad ...(ii)$$

By (i) & (ii)

a = 0.0045 moles

b = 0.00925 moles

$$\text{so } \%\ Fe_2O_3 = \frac{Wt(Fe_2O_3)}{\text{Total wt.}} \times 100 = \frac{b \times 160}{3} \times 100 = \frac{0.00925 \times 160}{3} \times 100 = 49.33\%.$$

$$\%\ Fe_2O_3 = \frac{Wt(Fe_2O_3)}{\text{Total wt.}} \times 100 = \frac{a \times 232}{3} \times 100 = \frac{0.0045 \times 232}{3} \times 100 = 34.8\,\%$$

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Illustration 25

Hydrogen peroxide solution (20 mL) reacts quantitatively with a solution of $KMnO_4$ (20 mL) acidified with dilute H_2SO_4. The same volume of $KMnO_4$ solution is just decolorized by 10 mL of $MnSO_4$ in neutral medium simultaneously forming a dark brown precipitate of hydrated MnO_2. The brown precipitate is dissolved in 10 mL of 0.2 M sodium oxalate under boiling condition in the presence of dilute H_2SO_4. Write the equations involved in the reactions and calculate the molarity of H_2O_2.

Solution:

Let molarity of H_2O_2 solution is M_1

20 ml, M_1 molar H_2O_2 reacts with 20 ml $KMnO_4$ in H_2SO_4 solution

$H_2O_2 + KMnO_4 + H_2SO_4 \rightarrow MnSO_4 + K_2SO_4 + H_2O + O_2$

no. of gram equivalent of H_2O_2 = no. of gram equivalent of $KMnO_4$

$(M \times n_f \times V)_{H_2O_2} = (M \times n_f \times V)_{KMnO_4}$

$$M_1 \times 2 \times 20 = M_{KMnO_4} \times 5 \times 20 \qquad ...(i)$$

20 ml $KMnO_4$ decolourise 10 ml $MnSO_4$ in neutral medium.

$KMnO_4 + MnSO_4 + H_2O \rightarrow MnO_2 + H_2SO_4 + KHSO_4$

no. of gram equivalent of $KMnO_4$ = no. of gram equivalent of MnO_2

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$M_{KMnO_4} \times 3 \times 20ml$ = no. of gram equivalent of MnO_2

$M_{KMnO_4} \times 3 \times 20 \times 10^{-3} = n_{MnO_2} \times 6/5$ ...(ii)

and obtained MnO_2 dissolved in 10 ml, 0.2 M $Na_2C_2O_4$

$MnO_2 + Na_2C_2O_4 + H_2SO_4 \rightarrow MnSO_4 + CO_2 + Na_2SO_4 + H_2O$

no. of gram equivalent of MnO_2 = no. of gram equivalent of $Na_2C_2O_4$

$n_{MnO_2} \times 2 = 0.2 \times 2 \times 10 \times 10^{-3}$

$n_{MnO_2} = 2 \times 10^{-3}$ moles ...(iii)

From equation (ii)

$$M_{KMnO_4} \times 3 \times 20 \times 10^{-3} = 2 \times 10^{-3} \times \frac{6}{5}$$

$$M_{KMnO_4} = \frac{2}{50}$$

From equation (1)

$$M_1 \times 2 \times 20 = \frac{2}{50} \times 5 \times 20$$

$M_1 = 0.1M$

Illustration 26

One litre of a mixture of O_2 and O_3 ($O_3 \longrightarrow O_2 + O^{2\ominus}$) at 1 atm and 273 K was allowed to react with an excess of acidified solution of KI. The iodine liberated required 40 ml of M/10 sodium thiosulphate solution for titration. What is the percent of ozone in the mixture ? Ultraviolet radiation of wave length 300 nm can decompose ozone. Assuming that one photon can decompose one ozone molecule, how many photons would have been required for the complete decomposition of ozone in the original mixture ?

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Solution:

Let in 1 litre mixture x litre is O_2 and y litre is O_3 at NTP

excess KI reduce O_3 into O_2 and $O^{2\ominus}$

$KI + O_3 + H_2O \rightarrow I_2 + O_2 + KOH$

no. of gram equivalent of O_3 = no. of gram equivalent of I_2

$(n \times n_f)_{O_3} = (n \times n_f)_{I_2}$

$$\frac{y}{22.4} \times 2 = n_{I_2} = n_{I_2} \times 2$$

$$n_{I_2} = \frac{y}{22.4} \times 2$$ ...(1)

Now, released I_2 required 40 ml, $\frac{1}{10}$M $Na_2S_2O_3$

$I_2 + Na_2S_2O_3 \rightarrow NaI + Na_2S_4O_6$

No. of gram equivalent of I_2 = no. of gram equivalent of $Na_2S_2O_3$

$$n_{I_2} \times 2 = \frac{1}{10} \times 1 \times 40 \times 10^{-3}$$

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$n_{I_2} = 2 \times 10^{-3}$ moles

Now from equation (1)

$$\frac{y}{22.4} = 2 \times 10^{-3}$$

$y = (44.8 \times 10^{-3})$ litre = 0.0448 litre

$x = 1 - y = 1 - 0.0448 = 0.9552$ litre

$$\% \text{ (by wt)} = \frac{wt(O_3)}{wt.(total)} \times 100$$

$$= \frac{n(O_3) \times 48}{n(O_3) \times 48 + n(O_2) \times 32} \times 100$$

$$= \frac{\frac{y}{22.4} \times 48}{\frac{y}{22.4} \times 48 + \frac{x}{22.4} \times 32} \times 100$$

$$= \frac{3y}{3y + 2x} \times 100$$

$$= \frac{3 \times 0.0448}{3 \times 0.048 + 2 \times 0.9552} \times 100$$

$$= \frac{0.1344}{2.0448} \times 100 = 6.57\ \%$$

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One photon required for breaking of one O_3 bond in to O_2 & O

$O_3 \xrightarrow{h\nu} O_2 + O$

So, no. of photon required = no. of O_3 bonds break

= no. of O_3 molecules

$= n_{O_3} \times N_A = \frac{0.0448}{22.4} \times 6 \times 10^{23} = 1.2 \times 10^{21}$ photons.

Illustration 27

A mixture of FeO and Fe_2O_3 is reacted with acidified $KMnO_4$ solution having a concentration of 2/5 M, 100 mL of which was used. The solution was then titrated with Zn dust which converted $Fe^{3\oplus}$ of the solution to $Fe^{2\oplus}$. The $Fe^{2\oplus}$ required 1000 mL of 2/15 M $K_2Cr_2O_7$ solution. Find the % by mol of FeO and Fe_2O_3.

Solution:

Let no. of moles of FeO is x millimoles & Fe_2O_3 is y millimoles in mixture.

$FeO + KMnO_4 + H_2SO_4 \rightarrow MnSO_4 + Fe_2(SO_4)_3 + K_2SO_4 + H_2O$

no. of milligram equivalent of FeO = no. of milli gram equivalent of $KMnO_4$.

$$x \times 1 = \frac{2}{5} \times 5 \times 100$$

x = 200 m moles.

Zn - dust convert all $Fe^{3\oplus}$ to $Fe^{2\oplus}$

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So, no. of millimoles of $Fe^{2\oplus}$ in solution is = (x + 2y).

$Fe^{2\oplus} + K_2Cr_2O_7 + H_2SO_4 \rightarrow Cr_2(SO_4)_3 + Fe^{3\oplus} + K_2SO_4 + H_2O$

no. of milli gram equivalent of $Fe^{2\oplus}$ = no. of milligram equivalent of $K_2Cr_2O_7$

$(x + 2y) \times 1 = \frac{2}{15} \times 6 \times 1000$

$(200 + 2y) = \frac{2}{15} \times 1000$

y = 300 millimoles.

$\%\ FeO\ (\text{by moles}) = \frac{x}{x+y} \times 100 = \frac{200}{500} \times 100 = 40\%$

$\%\ Fe_2O_3$ (by moles) = 60%

Illustration 28

A substance of crude copper is boiled in H_2SO_4 till all the copper has reacted. The impurities are inert to the acid. The SO_2 liberated in the action is passed into 100 mL of 0.4 M acidified $KMnO_4$($SO_2 \rightarrow SO_4^{2\ominus}$) The solution of $KMnO_4$ after passage of SO_2 is allowed to react with oxalic acid and requires 25 mL of 1 M oxalic acid. If the purity of copper is 95.25%, what was the weight of the sample.

Solution:

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Let given sample is w gram and no. of moles of Cu in given sample is x millimoles.

$Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$

x millimoles x millimoles

$SO_2 + KMnO_4 + H_2SO_4 \rightarrow MnSO_4 + H_2O + SO_3 + K_2SO_4$

no. of milligram eq. of SO_2 = no. of milligram eq. of $KMnO_4$

x × 2 = no. of gram eq. of $KMnO_4$

left $KMnO_4$ reduced by $H_2C_2O_4$

$KMnO_4 + H_2SO_4 + H_2C_2O_4 \rightarrow MnSO_4 + CO_2 + H_2O + K_2SO_4$

no. of milligram eq. of $KMnO_4$(left)

= no. of mill gram eq. of $H_2C_2O_4$

= 1 × 2 × 25

So, total milligram eq. of $KMnO_4$ = milligram eq. of SO_2 + milligram eq. of $H_2C_2O_4$

0.4 × 5 × 100 = x × 2 + 1 × 2 × 25

x = 75 millimoles

wt(Cu) = 95.25% of W

$75 \times 10^{-3} \times 63.5 = \frac{95.25}{100} \times W \Rightarrow W = 5$ gram.

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Illustration 29

H_2O_2 is reduced rapidly by $Sn^{2\oplus}$, the products being $Sn^{4\oplus}$ and water. H_2O_2 decomposes slowly at room temperature to yield O_2 and water. Calculate the volume of O_2 produced at 273 K and 1.00 atm when 200 g of 10.0 % by mass H_2O_2 in water is treated with 88.2 mL of 1 M $Sn^{2\oplus}$ and then the mixture is allowed to stand until no further reaction occurs.

Solution:

Let x millimoles of H_2O_2 react with 1M, 88.2 ml $Sn^{2\oplus}$

$H_2O_2 + Sn^{2\oplus} \rightarrow Sn^{4\oplus} + H_2O$

no. of milligram equivalent of H_2O_2

= no. of milligram eq. of $Sn^{2\oplus}$

$x_{millimole} \times 2 = 1 \times 2 \times 88.2$

x = 88.2 millimoles

Total given moles of H_2O_2 is

$$n_{total} = \frac{W(H_2O_2)}{34} = \frac{10\% \text{ of } 200 \text{ gram}}{34}$$

$$= \frac{20}{34} \text{moles}$$

= 0.5882 moles

= 588.2 millimoles.

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so. no. of moles of H_2O_2 that decomposed to O_2

= 588.2 – 88.2

= 500 millimoles

$$H_2O_2 \rightarrow H_2O + \frac{1}{2} O_2$$

$$n(O_2) \text{ released} = \frac{1}{2} \times 500 = 250 \text{ millimoles}$$

V_{O_2} (at 1atm, 273 K) = $250 \times 10^{-3} \times 22.4 = 5.6$ litre.

Illustration 30

0.84 g iron ore containing x percent of iron was taken in a solution containing all the iron in ferrous condition. The solution required x ml of a dichromatic solution for oxidizing the iron content to ferric state. Calculate the normality of dichromatic solution.

Solution:

Let Normality of $K_2Cr_2O_7$ solution is N

$Fe^{2\oplus} + K_2Cr_2O_7 + H^{\oplus} \rightarrow Fe^{3\oplus} + Cr^{3\oplus} + H_2O$

no. of gram eq. of $Fe^{2\oplus}$ = no. of gram eq. of $K_2Cr_2O_7$

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$n_{Fe^{2\oplus}} \times 1 = N \times x \times 10^{-3}$

$\frac{W_{Fe^{2\oplus}}}{56} \times 1 = N \times x \times 10^{-3}$

$\frac{\frac{x}{100} \times 0.84}{56} \times 1 = N \times x \times 10^{-3}$

N = 0.15 N

Illustration 31

5 g of pyrolusite (impure MnO_2) were heated with conc. HCl and Cl_2 evolved was passed through excess of KI solution. The iodine liberated required 40 mL of N/10 hypo solution. Find the % of MnO_2 in the pyrolusite.

Solution:

Let moles of MnO_2 in 5 gram sample is x.

$MnO_2 + HCl \rightarrow MnCl_2 + Cl_2 + H_2O$

no. of gram eq. of MnO_2 = no. of gram equivalent of Cl_2

$x \times 2 = n_{(Cl_2)} \times 2$

$x = n_{(Cl_2)}$

so $n_{I_2} = n_{Cl_2} = x$

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evolved Cl_2 passes through KI that released I_2

$Cl_2 + 2KI \rightarrow I_2 + 2KCl$

x moles x moles

Evolved I_2 required 40 ml, $\frac{N}{10}$ $Na_2S_3O_3$

$I_2 + 2Na_2S_2O_3 \rightarrow 2NaI + Na_2S_4O_6$

no. of gram equivalent of I_2 = no. of gram equivalent of hypo

$x \times 2 = \frac{1}{10} \times 40 \times 10^{-3}$

$x = 2 \times 10^{-3}$ moles

$\%\ MnO_2 = \frac{wt.(MnO_2)}{total\ wt.} \times 100$

$= \frac{x \times 87}{5} \times 100 = \frac{2 \times 10^{-3} \times 87}{5} \times 100$

= 3.48 %

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Illustration 32

A 458 g sample containing Mn_3O_4 was dissolved and all manganese was converted to $Mn^{2\oplus}$. In the presence of fluoride ion, $Mn^{2\oplus}$ is titrated with 3 lit of $KMnO_4$ solution (which was 1.25 N against oxalate in acidic medium), both reactants being converted to a complex of Mn(III) , What was the % of Mn_3O_4 in the sample ?

Solution:

$KMnO_4 + H_2C_2O_4 + H_2SO_4 \rightarrow CO_2 + MnSO_4 + K_2SO_4 + H_2O$

3 litre, M molar

Normality ($KMnO_4$) = Molarity × n_f

1.25 = M × 5

M = 0.25

Let x moles of Mn_3O_4 present in 458 gram sample

$Mn_3O_4 \rightarrow 3Mn^{2\oplus}$

x mole 3x moles

$Mn^{2\oplus} + KMnO_4 \rightarrow Mn^{3\oplus}$

no. of gram eq. of $Mn^{2\oplus}$ = no. of gram eq. $KMnO_4$

$3x \times 1 = \left(M \times n_f \times V\right)_{KMnO_4}$

3x × 1 = 0.25 × 4 × 3

x = 1 moles

$$\% \, Mn_3O_4 \text{ (by wt)} = \frac{w(Mn_3O_4)}{w_{total}} \times 100$$

$$= \frac{1 \times 229}{458} \times 100 = 50\%$$

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Illustration 33

80 gm of a sample of Anhydrous $CuSO_4$ was dissolved in water and made to 250 mL. 25 mL of this solution after taking usual precaution was treated with a little excess of KI solution. A white ppt. of Cu_2I_2 and iodine was evolved. The iodine so evolved required 40 mL of 1 M hypo solution. What is the approximate purity of $CuSO_4$ solution.

[Mol. wt. of $CuSO_4$ = 160]

Solution:

Let 80 gram sample contains x moles of $CuSO_4$ that dissolve in 250 ml solution

So 25 ml solution contain (x/10) moles $CuSO_4$

$CuSO_4 + KI \rightarrow Cu_2I_2 + I_2$

no. of gram eq. of $CuSO_4$ = no. of gram eq. of I_2

$$\frac{x}{10} \times 1 = n_{(I_2)} \times 2$$

$$n_{(I_2)} = \frac{x}{20}$$

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Evolved I_2 required 40 ml, 1M hypo solution

$I_2 + Na_2S_2O_3 \rightarrow NaI + Na_2S_4O_6$

no. of gram eq. of I_2 = no. of gram eq. of hypo

$n(I_2) \times 2 = 1 \times 1 \times 40 \times 10^{-3}$

$\frac{x}{20} \times 2 = 40 \times 10^{-3}$

$x = 40 \times 10^{-2}$ moles

% $CuSO_4$ (by mass) = $\frac{wt(CuSO_4)}{wt\ total} \times 100$

$= \frac{40 \times 10^{-2} \times 160}{80} \times 100 = 80\%$

Glossary:

Aliquot. A portion of the whole, usually a simple fraction. A portion of a sample withdrawn from a volumetric flask with a pipet is called an aliquot.

Analytical concentration. The total number of moles per litre of a solute regardless of any reactions that might occur when the solute dissolves. Used synonymously with formality.

Equivalent. The amount of a substance which furnishes or reacts with 1 mol of $H^{\oplus}$ ion (acid-base), 1 mol of electrons (redox), or 1 mol of a univalent cation (precipitation and complex formation).

Equivalent weight. The weight in grams of one equivalent of a substance.

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Equivalence point. The point in a titration where the number of equivalents of titrant is the same as the number of equivalents of analyte.

End point. The point in a titration where an indicator changes color.

Formula weight. The number of formula weights of all the atoms in the chemical formula of a substance.

Formality. The number of formula weights of solute per litre of solution; synonymous with analytical concentration.

Indicator. A chemical substance which exhibits different colors in the presence of excess analyte or titrant.

Normality. The number of equivalents of solute per litre of solution.

Primary standard. A substance available in a pure form or state of known purity which is used in standardizing a solution.

Standardization. The process by which the concentration of a solution is accurately ascertained.

Standard solution. A solution whose concentration has been accurately determined.

Titrant. The reagent (a standard solution) which is added from a buret to react with the analyte.

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